

Module 01

1. The following text is adapted from John Matheus's 1925 short story "Fog." A train has stopped at a station, where heavy fog has set in.

The little conductor stood on tiptoe in an effort to keep one hand on the signal rope, craning his neck in a vain and dissatisfied endeavor to pierce the miasma of the fog. The motorman chafed in his box, thinking of the drudging lot of the laboring man. He registered discontent.

Q. As used in the text, what does the word "registered" most nearly mean?

- A. circumvented
- B. matriculated
- C. esteemed
- D. evinced

Correct Answer: D

Domain: Craft and Structure

Subdomain: Words in Context

2. As a member of Indigenous Photograph, artist Tshepiso Mabula ka Ndogngeni (Xhosa) can ____ her work more broadly than she could without the organization's reach. \nPhotography editors around the world can search for Indigenous photographer members on the organization's website to find images that document and reflect the lives of Indigenous communities.

Q. Which choice completes the text with the most logical and precise word or phrase?

- A. Evacuate
- B. promote
- C. Transfigure
- D. Ascertain

Correct Answer: B

Domain: Craft and Structure

Subdomain: Words in Context

3. The committee's report was praised for its ____ tone: it criticized the policy's flaws without resorting to personal attacks. Instead, the authors acknowledged the program's original goals and then laid out evidence-based revisions that could make it more effective. They also anticipated likely objections from critics and addressed them directly, using careful qualifiers rather than sweeping claims. Even when describing costly failures, the report maintained a calm, controlled voice that made its recommendations sound credible rather than inflammatory.

Q.Which choice completes the text with the most logical and precise word or phrase?

- A bombastic
- B vitriolic
- C measured
- D capricious

Correct Answer: C

Domain: Craft and Structure

Subdomain: Words in Context

4. All stainless steel contains varying amounts of iron, carbon, and corrosion-inhibiting chromium. However, ferritic stainless steel, often used for induction cookers, contains a higher percentage of chromium (at least 10.5)% than does austenitic stainless steel as well as a higher concentration of iron. Unlike ferritic stainless steel, austenitic stainless steel has a face-centered cubic crystalline structure held stable by the presence of nickel and nitrogen. Austenitic stainless steel has two subtypes: the 200 series, often used for washing machines, and the 300 series, which has more nickel than the 200 series and is often used for storage containers or furnaces. Thus, stainless steel used to manufacture _____

Q.Which choice most logically completes th

A furnaces and stainless steel used to manufacture washing machines will both have a chromium content of less than .

B washing machines will have a higher concentration of nickel in its composition than stainless steel used to manufacture furnaces will.

C induction cookers will have a face-centered cubic crystalline structure, but stainless steel used to manufacture storage containers will not.

D storage containers and stainless steel used to manufacture induction cookers will have a similar concentration of nitrogen in their compositions.

Correct Answer: A

Domain: Information and Ideas

Subdomain: Inference

5. Biologists Rebecca M, Calisi-Rodriguez and George E, Bentley examined research on species such as dark-eyed juncos and tucos, which have both been studied under laboratory conditions as well as in the wild, to see whether there were significant differences between findings in the wild and in the lab. And, for example, they found for tucos that daytime is the most active period for wild individuals but not

for captive individuals. Calisi-Rodriguez and Bentley therefore concluded that the laboratory setting was likely _____.

Q. Which choice most logically completes the text?

A) more suitable than wild settings for studying tucos' patterns of rest and activity, since the controlled environment would presumably allow researchers to observe behavior more consistently and with fewer outside variables.

B) affecting the results for both tucos and dark-eyed juncos, meaning that laboratory conditions likely changed findings across multiple species in similar ways rather than influencing only one species.

C) more suitable for studying dark-eyed juncos than for studying tucos, implying that the laboratory environment produces more valid or more naturalistic behavioral results for juncos than it does for tucos.

D) interfering with tucos' patterns of rest and activity, such that captivity or lab conditions likely disrupted the animals' typical daily rhythms and led to activity patterns that differed from those observed in the wild.

Correct Answer: D

Domain: Information and Ideas

Subdomain: Inference

6 A road was recently built in a Maryland woodland that is home to the house finch. Some finches' nests were situated close to the new road and others were deeper in the woodland. Common ravens, which eat finch eggs, can spot nests near the open spaces of roads more easily than they can spot nests surrounded by woodland. Accordingly, researchers in Maryland trying to predict the impact of the new road on finches have suggested that _____.

Q. Which choice most logically completes the text?

A) finch eggs will make up about the same percentage of common ravens' diet as they did before the road was built, suggesting that the new open roadside areas will not meaningfully change ravens' ability to locate and raid finch nests.

B) the number of finches building nests near the road will gradually increase as the finches adapt to the presence of the common ravens, implying that finches will come to prefer roadside nesting despite the increased visibility of nests to predators.

C) finches nesting near the road will lose eggs to common ravens at a higher rate than finches nesting far from the road, because nests adjacent to the road are more exposed and therefore easier for ravens to detect than nests deeper in the woodland.

D) finches nesting far from the road will lay fewer eggs per nest than finches nesting near the road, implying that nest placement changes the number of eggs finches produce rather than primarily changing how vulnerable those eggs are to predation.

Correct Answer: C

Domain: Information and Ideas

Subdomain: Inference

7. Monthly Temperatures and Wing Centroid Sizes of Fruit Fly Specimens

Month	Average high (°F)	Average low (°F)	Average male wing centroid size (mm)	Average female wing centroid size (mm)
October	67	44	1.98	2.29
July	87	62	2.02	2.31
June	80	56	2.01	2.31
May	73	50	1.98	2.27

Drosophila (fruit flies) have generation times of 10-12 days, so seasonal changes in humidity and other environmental conditions can drive seasonal fluctuations in chromosome rearrangements in species such as *D. persimilis* and *D. mediopunctata*. *Drosophila* body size (for which wing centroid size serves as a proxy measure) correlates with life span. Banu Sebnem Önder and Cansu Fidan Aksoy measured the wing sizes of members of a *D. melanogaster* population in Yesilöz, Turkey, that were collected monthly between May and October over three years. Their research suggests that *Drosophila* collected in relatively warmer months should tend to have a longer life span, as is illustrated by the finding that ----

Q. Which choice most effectively uses data from the table to complete the assertion?

A) the average monthly low temperature was higher in June than in May, a comparison that describes one part of the climate pattern but does not, by itself, connect temperature to the body-size proxy discussed in the passage.

B) the average male wing centroid size was larger in July than in October, which uses the table's wing-size data and provides a concrete example of how specimens from one set of months differ in the body-size proxy highlighted by the researchers.

C) the average female wing centroid size was 2.02 mm in July but was 2.31 mm in June, a statement that gives two precise measurements but (as written) does not align cleanly with the claim that warmer-month specimens should tend to show the life-span-associated pattern.

D) the average female wing centroid size was consistently larger than the average male wing centroid size in all four months in the table, a trend about sex differences that does not directly illustrate the temperature-linked pattern the sentence is trying to support.

Correct Answer: C

Domain: Information and Ideas

Subdomain: Command of evidence

8. Neighboring Species and Target Species in Various Locations

Neighboring species	Target species	Location
musk thistle	Peruvian lily	Argentina
creeping thistle	wild radish	United Kingdom
leafy spurge	yellow evening primrose	United States

In a study of interactions between plants, researchers Carolina Laura Morales and Anna gathered data about flowering plants growing alongside each other in various locations. In each case, the researchers identified one plant as a "target species" and a plant as a "neighboring species."

An example of a neighboring species is the musk which grows alongside the Peruvian lily in _____

Q.Which choice most effectively uses data from the table to complete the example?

- A the United States.
- B Japan.
- C the United Kingdom.
- D Argentina.

Correct Answer: D

Domain: Information and Ideas

Subdomain:Command of evidence

9.The following text is from George Eliot's 1857 short story "The Sad Fortunes of the Rev. Amos Barton." In the text, the narrator addresses the reader directly and alludes to a discussion among Rev. Amos Barton's neighbors.

It was fortunate for the Rev. Amos Barton that he did not, like us, overhear the conversation recorded in the last chapter. Indeed, what mortal is there of us, who would find his satisfaction enhanced by an opportunity of comparing the picture he presents to himself of his own doings, with the picture they make on the mental retina of his neighbours? We are poor plants buoyed up by the air-vessels of our own conceit: alas for us, if we get a few pinches that empty us of that windy self-subsistence! The very capacity for good would go out of us.

Q.Which choice best states the main idea of the text?

- A) Although people grasp the importance of honesty, they typically resist confronting others about their flaws, preferring to avoid uncomfortable truth-telling even when they believe frankness would be morally better.
- B) Although people wish to be seen as considerate, the slightest setbacks will often discourage them from being so, since even minor frustrations can drain their patience and reduce their willingness to act kindly.
- C) People tend to fixate more often than they should on whether their acquaintances think highly of them, repeatedly worrying about approval and reputation even when such concerns are unproductive.
- D) People are better off not knowing about the discrepancy between their own self-image and what others think of them, because overhearing others' judgments would puncture their self-satisfaction and leave them less confident and less capable of sustaining their good intentions.

Correct Answer: D

Domain: Craft and Structure

Subdomain: Central Ideas and Details

10. Roy McLendon's *Moonlit St. Lucie*, a riverscape featuring the silhouette of a single palm tree against the backdrop of shimmering water and a brilliant moonlit sky, is typical of paintings by the Florida Highwaymen, an informal collective of landscape artists mainly active in the 1950s and '60s. Remarkable for anticipating and amplifying cultural perceptions of Florida that became pervasive in the public consciousness, paintings by the Highwaymen are readily identifiable by the natural iconography—placid inland rivers, windswept palm trees—that McLendon and colleagues perpetually revisited.

Q. Which choice best states the main idea of the text?

A Although similar in its subject matter to many paintings by the Florida Highwaymen, *Moonlit St. Lucie* is now more highly regarded than other Florida Highwaymen paintings are.

B Representative images found across many paintings by McLendon and other Florida Highwaymen came to be widely associated with Florida in part due to the Florida Highwaymen's influence.

C Although paintings by the Florida Highwaymen were once celebrated for their depictions of Florida's natural environments, the popularity of these paintings waned after the 1960s.

D The placid inland rivers and windswept palm trees that are typical of McLendon's works, which are otherwise indistinguishable from other Florida Highwaymen paintings, help to differentiate McLendon's paintings from those of his colleagues.

Correct Answer: B

Domain: Craft and Structure

Subdomain: Central Ideas and Details

11. Superlubricity, the state of virtually no friction between materials, has desirable applications in many industries. For example, it can make aircraft engines more efficient. To produce a coating that achieves superlubricity, Chanaka Kumara and colleagues broke down carbon nanotubes into fragments of graphene to fully cover two surfaces that would rub together. Friction between pieces of graphene is generally extremely low, and when the researchers added a drop of oil as lubrication, that friction nearly vanished. This new coating may drastically lower friction-related energy costs.

Q. According to the text, what happened when the researchers added oil to the surfaces covered in graphene fragments?

A) All the pieces of graphene collected on just one of the two surfaces, leaving the other surface incompletely coated and implying that the researchers' attempt to achieve full coverage failed because the fragments clumped together.

B) The low amount of friction between the surfaces became even lower, dropping further once the oil was added so that the motion between the graphene-covered surfaces approached a near-frictionless state.

C) Carbon nanotubes on the surfaces fractured into smaller pieces, suggesting that the key experimental result occurred when the nanotubes broke apart rather than when the oil was introduced as lubrication.

D) Friction between the surfaces did not noticeably change right away, remaining approximately the same even after the oil was added, which would contradict the passage's statement that the friction "nearly vanished" after lubrication.

Correct Answer: B

Domain: Craft and Structure

Subdomain: Central Ideas and Details

12 Known for the albums *Someday My Prince Will Come* and *Milestones*, jazz trumpeter Miles Davis collaborated several times with pianist Gil Evans. Their 1958 adaptation of George Gershwin's opera *Porgy and Bess* bears little resemblance to the 1935 original. Davis and Evans felt no desire to please listeners expecting an exact duplication of the opera. They omitted parts, such as the aria "I Got Plenty of Nuthin'," and sometimes made only brief gestures toward Gershwin's melodies. But Davis and Evans's willingness to recompose Gershwin's work led to one of the most enduring albums in Davis's catalog.

Q. Which choice best describes the function of the underlined sentence in the text as a whole?

A It presents examples to support a claim made earlier in the text.

B It shows how two artists benefited from ignoring certain conventions.

- C It proposes a reason why one work of art is widely thought to be more successful than another.
- D It undermines an assertion made later in the text.

Correct Answer: A

Domain: Craft and Structure

Subdomain: Rhetorical Structure and Purpose

13. Scholarly interest in literary juvenilia-writings by children and teenagers-tends to focus on unpublished works by authors who became famous as adults, such as W.H. Auden's poem "Autumn," which he wrote around age 15, because they offer insights into their authors' artistic development. But some scholars also argue that recovering juvenilia by lesser-known writers is essential to understanding literary history: Daisy Ashford's novels, which she published as a child, were widely read by contemporaries and are therefore deserving of closer attention.

Q.Which choice best states the main purpose of the text?

- A To describe the challenges famous writers encountered when seeking to publish works written in their childhood
- B To argue that Ashford's novels have more literary merit than Auden's juvenilia do
- C To compare the accomplishments of young writers with those of their adult contemporaries
- D To present reasons why literary scholars consider juvenilia to be valuable resources

Correct Answer: D

Domain: Craft and Structure

Subdomain: Purpose

14. Researchers César A. Hidalgo, Elisa Castañer, and Andres Sevtsuk created a computer model to predict the mix of businesses and places of interest found in a given neighborhood. The team used data from the Google Places API service to help identify florists, beauty salons, and other businesses and map their locations. This approach has some limits - data from Places API tend to be restricted to places that are customer facing - but the data set nonetheless provides an extremely reliable source to study colocation patterns of neighborhood amenities.

Which choice best describes the function of the underlined portion in the text as a whole?

- A It describes the imprecise and subjective nature of neighborhood boundaries.

B It introduces a model that a research team built to evaluate the mix of amenities in urban neighborhoods.

C it emphasizes the potential utility of the team's model.

D It identifies a specific flaw in using a data set about amenities in cities.

Correct Answer: D

Domain: Craft and Structure

Subdomain: Rhetorical Structure and Purpose

15. The following text is from Sarah Helen Whitman's 1878 poem "The River's Song."

Ah, pure joy to see the gleaming tide

Rejoicing sweep across the tumbled stones;

Often checked, but never ceasing flow—

Moved by a force that will not accept rest;

To watch the brilliant tints of sun and cloud.

The drifting gentle mist, floats ghost-like,

Into the quiet air, to feel the calm

Magic of the stream soothe all the mind,

Healing the weary thought, making the spirit

Sing happy as its own clear, running waters!

Which choice best describes the function of the underlined portion in the text as a whole?

A It depicts the landscape as a frightening and overwhelming sight.

B It defines the river's current as a persistent and unyielding momentum.

- C It reveals the speaker's uncertainty about the natural environment.
- D It establishes a difference between the river's current and the speaker's emotions.

Correct Answer: B

Domain: Craft and Structure

Subdomain: Rhetorical Structure and Purpose

16. For many years, scientists assumed that Jupiter's moon Europa was the only place beyond Earth in the solar system where large-scale ice quakes—powerful tremors produced by the sudden shifting of enormous slabs of frozen water—could generate towering waves. In a study intended to expand our understanding of such wave-forming events, planetary geologist Lena Ortiz and her team published findings documenting a gigantic ice slide, leaving behind a trail of disruption so extensive that it could only have resulted from a tremendous mass of moving water, on Saturn's moon Enceladus.

Q. According to the text, why was Ortiz and her team's research on the Enceladus ice slide significant?

- A The evidence collected by Ortiz and her team supports the idea that powerful, wave-generating events can occur on a body other than Earth.
- B The evidence collected by Ortiz and her team shows that Earth once experienced more intense ice quakes than it does now.
- C The team's findings help scientists argue that Enceladus is more likely to support human habitats than was once assumed.
- D The team's findings enable geologists to develop a precise timeline of how Enceladus's surface has changed over time.

Correct Answer: A

Domain: Information and Ideas

Subdomain: Central Ideas and Details

17. In 1921, biologist Alice M. Thornton presented experimental data suggesting that a mysterious livestock illness was spread from infected ticks to healthy cattle. For several years, other scientists dismissed Thornton's explanation, until she finally persuaded Dr. James Cooper, the lead veterinarian at a major agricultural research station in Kansas, to examine her claim. Cooper selected a group of healthy cows that were deliberately exposed to infected ticks and then had their blood drawn, trapping the infectious agent in vials. By injecting this blood into another group of previously healthy cows, Cooper was able to demonstrate a successful transmission of the illness—the first clear proof that the disease was carried by infected ticks.

Q.Which choice best states the main idea of the text?

- A It's difficult to control the spread of livestock illnesses because they often devastate rural economies.
- B Cooper's experiment offered strong support for another researcher's theory about how a disease is transmitted.
- C Cooper invented a groundbreaking technique for examining the symptoms of unfamiliar animal diseases.
- D Blood testing is the single most important diagnostic tool for treating livestock illnesses carried by ticks.

Correct Answer: B

Domain: Craft and Structure

Subdomain: Central Ideas and Details

18. *To Kill a Mockingbird* is a 1960 novel by Harper Lee. In the novel, the narrator Scout describes events involving her father, Atticus Finch, and the residents of Maycomb, Alabama. The novel contrasts Atticus's strong moral principles with the prejudice and unfairness shown by many of his neighbors, such as when Atticus says _____

Q.Which quotation from *To Kill a Mockingbird* most effectively illustrates the claim?

- A "You never really understand a person until you consider things from his point of view... until you climb into his skin and walk around in it."
- B "Maycomb was an old town, but it was a tired old town when I first knew it."
- C "Until I feared I would lose it, I never loved to read. One does not love breathing."
- D "People in their right minds never take pride in their talents."

Correct Answer: A

Domain: Information and Ideas

Subdomain: Command of Evidence

19. Estimates of Pacific Sleeper Shark Lifespan

Study	Year	Estimation method	Approximate lifespan (years)
Early marine ecologist	1940	Tagging individuals and tracking yearly growth	350
Ramirez	2010	Satellite tagging and measurement of growth	350
Lee et al.	2017	Carbon dating and analysis of eye tissue	350–500
Patel et al.	2023	Study of metabolic rate and feeding behavior	400

The Pacific sleeper shark is a slow-moving deep-sea shark that inhabits the North Pacific Ocean. Because this animal is rarely seen and difficult to observe directly, its lifespan has been challenging to determine. Nevertheless, estimates of its longevity have shown striking agreement over time; deep-sea biologist Jordan McClure argues that, drawing on information from other well-studied cold-water sharks, researchers using a variety of techniques are likely to arrive at similar approximations of the Pacific sleeper shark's lifespan.

Q. Which choice best describes data from the table that support McClure's argument?

A The approximate lifespans determined by the early marine ecologist, Ramirez, and Patel et al. all fell within the approximated lifespan range determined by Lee et al., even though there were differences in the methods used by each of the four groups.

B The study by the early marine ecologist used tagging for measurements and produced the lowest approximated lifespan, while the study by Lee et al. used carbon dating and produced the highest approximated lifespan.

C In their study, Patel et al. used metabolic rate and feeding behavior to determine the Pacific sleeper shark's lifespan approximation to be 400 years.

D Though the lifespans approximated by the early marine ecologist and Ramirez were identical to each other, the lifespans approximated by Patel et al. and Lee et al. differed significantly from any other approximation.

Answer A

Domain Information & Idea

Sub Domain Command of Evidence

20. Although now known for eating almost nothing but seeds, wild turkeys once had a much broader menu. Researchers estimate that thousands of years ago, turkeys began relying primarily on seeds, but before that they likely ate a mixture of plants and insects. To explore this idea, Dr. Lena Ortiz and her team at the National Wildlife Research Center compared fossilized turkey beaks with beaks from present-day turkeys. Noting that the fossilized beaks showed a wider range of tiny grooves and wear patterns than modern beaks do, Ortiz proposed that a greater variety of grooves and wear patterns reflects a greater variety of foods eaten; this would mean that ancient turkeys did in fact have a more diverse diet than modern turkeys.

Q. Which finding, if true, would most directly undermine Ortiz's hypothesis?

A Some beak samples included in the study were confirmed to be from modern turkeys and therefore only showed grooves and wear patterns associated with seed eating.

B Several beak samples not included in the study showed that modern turkey beaks can display the same range of grooves and wear patterns as ancient beaks do, depending on the types of seeds consumed.

C Several beak samples not included in the study suggested that ancient turkeys adapted over time to changes in the availability of food sources in their environment.

D Some beak samples included in the study were from a bird species closely related to turkeys and contained grooves and wear patterns associated with a mixed diet.

Answer B

Domain Information & Idea

Sub Domain Command of Evidence

21. In the deserts of the southwestern United States, the *Noctanis audax*—sometimes called the lantern-eyed spider—is notable for its unconventional hunting strategy. Behavioral ecologist Lena Carraway ran a series of experiments to determine how this spider, which does not depend on a traditional stationary web to sense nearby prey as many spiders do, uses its body to detect and leap onto

insects; subsequent trials suggested that *Noctanis audax* must still rely on some kind of vibration, even if not transmitted through a web. Carraway and her colleagues propose that the lantern-eyed spider not only uses its oversized eyes to visually track prey but also depends on subtle vibrations detected through sensory hairs on its legs to “hear” potential prey, whether it is on the ground or gliding through the air.

Q. Which finding, if true, would most directly support Carraway’s hypothesis?

A *Noctanis audax* can continue to catch prey when its eyes are covered, but only if it has spun a dense web to trap insects.

B Other species closely related to *Noctanis audax* are unable to hunt effectively without webs, even though they share similarly large eyes and leg structures.

C Although all *Noctanis audax* individuals have large eyes and vibration-sensitive leg hairs, some spin small webs in especially cluttered habitats such as thick sagebrush stands.

D Specimens of *Noctanis audax* were able to catch flying insects inside a smooth, webless chamber even when specimens of several other spider species were unable to capture any prey there.

Answer D

Domain Information & Idea

Sub Domain Command of Evidence

22. Hydrilla, an invasive aquatic plant originally from parts of Asia, first appeared in Florida canals in the 1960s and then spread throughout lakes and rivers in the southeastern United States after being transported accidentally on the trailers of recreational fishing boats. Recent studies comparing the condition of many lakes and reservoirs in regions with heavy human use, such as marinas and water diversion projects, to that of relatively undisturbed wetlands and headwater streams have shown increasing ecological disruption from hydrilla growth in the areas with more human activity. Thus, researchers can conclude that _____

Q. Which choice most logically completes the text?

A the hydrilla populations in heavily used waterways have likely evolved more quickly than those in relatively undisturbed wetlands and streams.

B the condition of human-impacted lakes and reservoirs was more similar to that of relatively undisturbed wetlands and streams in the past than it is today.

C the spread of hydrilla has been aided by human activity and has damaged the aquatic ecosystems in which that activity has occurred.

D aquatic organisms living in relatively undisturbed ecosystems have not been exposed to hydrilla.

Answer C

Domain Information & Idea

Sub Domain Inference

23. Ratified by the United States and the Republic of Texas in the mid-1840s, a commercial agreement between the two governments reduced taxes on items shipped from Texas into the United States, making it far cheaper for merchants to sell Texan goods in US markets. At the same time, the agreement granted US-made goods reduced tariffs when entering Texas, giving a clear advantage to American products there. Furthermore, the agreement allowed the United States to establish an exclusive naval coaling and repair station along a key section of the Texas coastline. While many leaders in Texas interpreted these terms as evidence that the United States respected Texas's status as an independent nation, the agreement in practice had the effect of _____

Q. Which choice most logically completes the text?

A promoting a substantial rise in both American influence over Texas's economy and American activity along the Texan coast.

B discouraging Texan merchants from investing in local enterprises because they preferred the supposedly safer financial opportunities available in eastern US cities.

C broadening the financial advantages that Texas enjoyed after relinquishing its sovereignty to outside powers.

D restricting the volume of trade that could be legally arranged with nations that had not entered into a comparable agreement with Texas.

Answer A

Domain Information & Idea

Sub Domain Inference

24. In 1910, American mountaineers Clara Hastings, Daniel Price, and Leo Morton attempted to scale the highest peak in Alaska. Unfortunately, the _____ effort to descend safely from the mountain.

Q. Which choice completes the text so that it conforms to the conventions of Standard English?

A climbs' failure led to the mountaineers'

B climb's failure led to the mountaineers'

C climb's failure led to the mountaineers

D climbs failure led to the mountaineer's

Answer B

Domain Standard English Convention

Sub Domain Punctuation

25. Unlike those of the pale-spotted woodland moth, _____ they helped the moth remain hidden on tree trunks darkened by soot near growing factory towns.

Q. Which choice completes the text so that it conforms to the conventions of Standard English?

A the dark-winged woodland moth's markings offered a benefit during the height of industrial expansion:

B a benefit during the height of industrial expansion was the dark-winged woodland moth's markings:

C the dark-winged woodland moth has markings that offered a benefit during the height of industrial expansion:

D the dark-winged woodland moth had a benefit during the height of industrial expansion because of its markings:

Answer A

Domain Standard English Convention

Sub Domain Modifier

26. Certain hormones are associated with sleep patterns and mood regulation. Researchers examined a specific gene and discovered that some individuals have the amino acid valine at a particular position instead of the amino acid serine. Individuals with serine at that position generally show average levels of sleep disturbances and mood swings. _____ individuals with valine at that position tend to show fewer sleep disturbances and milder mood swings.

Q. Which choice completes the text with the most logical transition?

A Indeed,

B Moreover,

C Accordingly,

D Nonetheless,

Answer D

Domain Expression of Idea

Sub Domain Transition

27. While researching a topic, a student has taken the following notes:

- The factors that affect seed germination rates have been well studied in rainforest plants but not in desert plants.
- A team led by Dr. Elena Rodriguez of the University of Arizona investigated which factors influence desert plant seed germination.
- Rodriguez's team obtained germination and climate data for over 2,500 plant species and analyzed the data with statistical models.
- Higher germination rates were associated with environments having unpredictable rainfall patterns.
- Desert plants may exhibit higher germination rates to capitalize on brief periods of moisture.

Q. The student wants to emphasize the aim of the research study. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A Researchers wanted to determine which factors affect desert plant seed germination because such factors have been well studied in rainforest plants but not in desert plants.

B After they collected data for over 2,500 plant species, researchers found that higher germination rates were associated with environments having unpredictable rainfall patterns.

C We now understand that desert plants may exhibit higher germination rates to capitalize on brief periods of moisture.

D Researchers gathered germination and climate data for over 2,500 plant species and analyzed the data with statistical models.

Answer A

Domain Expression of Idea

Sub Domain Rhetorical Synthesis

Module - 2 (5-M-2)

1. Possessing an outstanding collection of public art, Chicago has everything from monumental sculptures like Jean Dubuffet's *Monument with Standing Beast* at sites like James R. Thompson Center to innovative street art like Justus Roe's mural *Alpha Omega* located on North Ashland Avenue. The ____ public art on display in the city can thus satisfy any art lover.

Which choice completes the text with the most logical and precise word or phrase?

- A. breadth of
- B. confusion about
- C. controversy over
- D. apathy toward

Craft & Structure

Words in Context

Ans A

2. In a 2018 article about films depicting the experiences of Black Americans, critics for the *New York Times* praise William Greaves's 1968 film *Symbiopsychotaxiplasm, Take One* as "a vital artifact of its time" and Gordon Parks's 1969 film *The Learning Tree* as "richly textured." Fans of the two films hope that such ____ will attract new audiences to these works.

Which choice completes the text with the most logical and precise word or phrase?

- A. foresight
- B. acclaim
- C. impartiality
- D. ambivalence

Craft & Structure

Words in Context

Ans B

3. The fossil remains of the individual known as LH 4, discovered in Tanzania in 1974, can help paleoanthropologists not only ____ steps in the evolution of hominids but also illuminate the Pliocene epoch generally, revealing important details about the time in which LH 4 lived.

Which choice completes the text with the most logical and precise word or phrase?

- A. prioritize
- B. yield
- C. exploit
- D. discern

Craft & Structure

Words in Context

Ans D

4. Companies are providing consumers with more opportunities to purchase customized products than ever before. Whether buying customized sneakers, jewelry, or clothing, consumers can participate in the design of products to meet their specific needs and tastes. In turn, companies profit too: studies have shown that consumers are willing to pay more and wait longer for a customized product. Still, it can be difficult for companies to offer customization while keeping costs low, as the standard methods of mass production may not be able to accommodate making a unique product each time.

Which choice best describes the overall structure of the text?

- A. It discusses several recent innovations in product manufacturing and then suggests some potential applications of those innovations.
- B. It describes a company's recent success with new products and then explains multiple factors that may have contributed to that success.
- C. It presents two contrasting product-marketing techniques and then provides examples of one of those techniques.
- D. It introduces a trend in consumer products and then explains how the trend both benefits and poses a challenge to companies.

Craft & Structure

Text structure & purpose

Ans D

5. *The Hundred Days*, first published in 1998, is a novel in Patrick O'Brian's Aubrey/Maturin series, which includes twenty completed books. Some critics have found fault with the abrupt endings of *The Hundred Days* and other books in the series, saying that they do not finish conclusively but arbitrarily stop. But other critics argue that the books should not be thought of as discrete texts with traditional beginnings and endings but as a single incredibly long work, similar to other multivolume stories, such as Marcel Proust's *In Search of Lost Time*.

Which choice best describes the function of the underlined portion in the text as a whole?

- A. It presents a reason most critics think the Aubrey/Maturin series should have the literary renown of similar works like *In Search of Lost Time*.
- B. It explains why many critics find the Aubrey/Maturin novels to be entertaining despite flaws in the novels' structures.

C. It argues that the unusual structure that O'Brian uses for *The Hundred Days* makes it one of his most entertaining books.

D. It summarizes a positive assessment of a particular characteristic of the Aubrey/Maturin novels.

Craft & Structure

Text structure & purpose

Ans D

6.Text 1

In separate studies, Stephen Meding and Robert J. Zasoski and Sanaá Wahbi and colleagues examined whether plants transfer nutrients to one another using a common mycorrhizal network (CMN)—a lattice of fungal strands in the soil. Meding and Zasoski excluded all pathways other than the CMN by using barriers to keep the plants' root systems separate while allowing mycorrhizal strands through—a crucial step Wahbi and colleagues' study did not take.

Text 2

Meding and Zasoski took the necessary precaution of separating the plants' root systems (thereby excluding root-to-root transmission). However, any barrier used must allow the thread-like hyphae of a CMN to pass through, and this permeability would also allow liquids through. Thus, the researchers' experimental design cannot ensure that any nutrient transfer observed can be attributed to a CMN and not to some other pathway.

Based on the texts, the author of Text 1 and the author of Text 2 would most likely agree with which statement?

A. The barriers used in Meding and Zasoski's study effectively excluded root-to-root transmission of nutrients.

B. Excluding root-to-root transfer of nutrients between plants is sufficient to ensure that any observed nutrient transfer must involve a CMN.

C. Wahbi and colleagues' study did not find convincing evidence of nutrient transfer between individual plants.

D. A barrier that is impervious to both roots and fungal strands is necessary to evaluate nutrient transfer via a CMN.

Craft & Structure

Cross text Connection

Ans A

7. Optimal foraging theory (OFT) holds that animals' foraging behaviors reflect cost-benefit trade-offs that vary by species and with dynamic ecological circumstances. One such circumstance is lunar intensity, which Joel Brown and colleagues found to be negatively associated with foraging by Arizona pocket mice but Deborah J. Curtis and colleagues found to be positively associated with foraging by mongoose lemurs. This discrepancy is explicable in terms of OFT: the lemurs' greater reliance on vision means that higher lunar intensity benefits them more than it benefits the mice.

Which choice best describes the finding made by Curtis and colleagues, as presented in the text?

- A. As lunar intensity increases, Arizona pocket mice reduce their foraging activity.
- B. As lunar intensity increases, mongoose lemurs increase their foraging activity.
- C. During periods of high lunar intensity, mongoose lemurs increase their reliance on vision.
- D. Increased lunar intensity is more beneficial for mongoose lemurs than it is for Arizona pocket mice.

Information & Idea

Detail

Ans B

8. Scientists studying coral reefs have long known that rising ocean temperatures can cause coral bleaching, a process in which corals expel the microscopic algae that provide much of their energy. Recently, marine biologist Elena Ramirez and her team investigated whether certain reefs recover more quickly because of differences in the surrounding fish populations. After monitoring reefs across three tropical regions for several years, the researchers found that reefs with a greater diversity of grazing fish regained healthy algae more rapidly than reefs with fewer such species. To corroborate this finding, the team compared satellite imagery with underwater surveys and discovered a remarkably consistent pattern across locations. Although the researchers caution that additional studies are needed to establish a direct causal relationship, their results suggest that protecting fish biodiversity may enhance the resilience of coral ecosystems as oceans continue to warm.

Which choice best describes the function of the underlined sentence in the text as a whole?

A. It provides additional evidence that corroborates the researchers' primary finding by showing that two independent methods produced consistent results.

B. It introduces a competing hypothesis suggesting that satellite imagery is more reliable than underwater surveys for studying coral reefs.

C. It emphasizes the researchers' uncertainty by explaining why the study's findings should be considered inconclusive despite extensive data collection.

D. It shifts the discussion from coral bleaching to the economic benefits of conserving marine ecosystems, expanding the scope of the argument.

Craft & structure

Text Structure & Purpose

Ans A

9. The bird species *Hypocnemis cantator* (the Guianan warbling antbird) shares some territory in French Guiana with *Thamnomanes caesius* (the cinereous antshrike), which emits a loud alarm call when it detects predators. Biologist Ari Martínez and colleagues recorded *T. caesius* alarm calls and played them in the vicinity of wild *H. cantator*. Finding that the birds often froze in place or scattered into vegetation upon hearing the calls, they concluded that *H. cantator* associates *T. caesius* alarm calls with danger.

Which finding, if true, would most directly support Martínez and colleagues' conclusion?

A. In some instances, *H. cantator* froze in place or scattered into vegetation when Martínez and colleagues approached but before they began playing sounds.

B. Other bird species than *H. cantator* also showed a tendency to freeze in place or scatter into vegetation when Martínez and colleagues played different *T. caesius* alarm calls.

C. Martínez and colleagues played alarm calls from different *T. caesius* individuals and observed no significant variation in the responses of *H. cantator*.

D. When Martínez and colleagues played control sounds of random noise in the vicinity of *H. cantator*, the birds displayed no reaction.

Information & Idea

Command of evidence Textual

Ans D

10.

Summit	Elevation (meters)	Mountain range	Prominence (meters)
Kangto	7,060	Assam Himalaya	2,195
Saser Kangri III	7,495	Saser Karakoram	850
Langpo	6,965	Sikkim Himalaya	560
Sri Kailash	6,932	Garhwal Himalaya	1,092
Mount Lakshmi	6,983	Rimo Karakoram	800

Mountain summits are often described in terms of their elevation, or height above sea level. But a summit's elevation may not be as good an indication of how high the mountain appears to observers as is the summit's prominence, or its height above its surroundings, and these values can differ significantly. For example, the Indian mountain of ____.

Which choice most effectively uses data from the table to complete the example?

- A. Saser Kangri III has an elevation of 7,495 meters but a considerably lower prominence of 850 meters.
- B. Kangto has a much higher prominence than does Langpo.
- C. Kangto has a high prominence but is from a different mountain range than Mount Lakshmi, which has a lower prominence.
- D. Sri Kailash has an elevation of 6,932 meters and is considered the highest mountain from the Garhwal Himalaya range.

Information & idea

Command of Evidence

Ans A

11. *The Underdogs* is a 1915 novel by Mariano Azuela, originally written in Spanish. In the novel, Azuela makes a comparison using natural imagery to convey the mental state of the protagonist, Demetrio: ____

Which quotation from a translation of *The Underdogs* most effectively illustrates the claim?

- A. The river crept along, murmuring as the waters rose and fell in small cascades.
- B. Demetrio suppressed a sigh. Memories crowded and buzzed through his brain like bees about a hive.
- C. When he reached the summit, he glanced down to see the sun steeping the valley in a lake of gold.
- D. Demetrio laid a hand on his hair which covered his ears, and stretching himself out for a long time, seemed to be lost in thought.

Information & idea

Command of Evidence

Ans B

12. The linguistic niche hypothesis (LNH) posits that the exotericity of languages (how prevalent non-native speakers are) and grammatical complexity are inversely related, which the LNH ascribes to attrition of complex grammatical rules as more non-native speakers adopt the language but fail to acquire those rules. Focusing on two characteristics that are positive indices of grammatical complexity, fusion (when new phonemes arise from the merger of previously distinct ones) and informativity (languages' capacity for meaningful variation), Olena Shcherbakova and colleagues conducted a quantitative analysis for more than 1,300 languages and claim the outcome is inconsistent with the LNH.

Which finding, if true, would most directly support Shcherbakova and colleagues' claim?

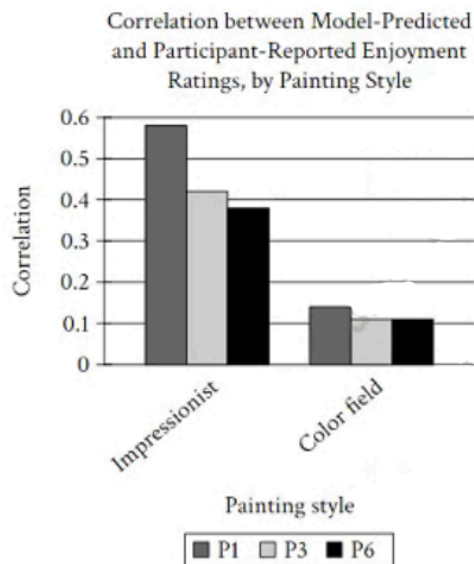
- A. Shcherbakova and colleagues' analysis showed a slightly positive correlation between grammatical complexity and fusion.
- B. Shcherbakova and colleagues' analysis showed a slightly negative correlation between grammatical complexity and exotericity.
- C. Shcherbakova and colleagues' analysis showed a slightly positive correlation between fusion and exotericity and between informativity and exotericity.
- D. Shcherbakova and colleagues' analysis showed a slightly negative correlation between grammatical complexity and fusion and between grammatical complexity and informativity.

Information & idea

Command of Evidence

Ans C

13.



Neuroscientist Kiyohito Iigaya and colleagues developed a computational model to predict how much a person will enjoy a particular work of art on a scale from 1 (not at all) to 4 (very much). They then recruited participants to use the same scale to rate several sets of paintings in various styles and used the model to make predictions for each participant. The figure shows how closely the model's predictions corresponded to the participants' self-reported enjoyment.

Which choice most effectively uses data from the graph to complete the statement?

P1 would derive _____.

- A. P1 would derive more aesthetic pleasure from impressionist paintings than from color-field paintings.
- B. P1 would derive less aesthetic pleasure from impressionist paintings than from color-field paintings.
- C. P1's ratings for impressionist and color-field paintings would equal one another.
- D. P1's ratings for impressionist and color-field paintings would differ from one another

Information & idea

Command of Evidence

Ans A

14. Arthurian legends (tales related to the character of King Arthur) derive from many often contradictory sources, such as *Annales Cambriae*, composed around 970, and *Tom a Lincoln* from around 1607. Sir Thomas Malory's 15th-century text *Le Morte d'Arthur* was an attempt to compile these stories into a coherent narrative. Many of Malory's sources derive from Geoffrey of Monmouth's *History of the Kings of Britain*, written in the 1130s. While neither *History* nor any works that predate it mention Arthur's famous Round Table at which his knights assembled, *Le Morte d'Arthur* does, suggesting that _____

A. when a version of an Arthurian legend contradicted the version in *History*, Malory preferred to include Geoffrey of Monmouth's version in *Le Morte d'Arthur*.

B. *Le Morte d'Arthur* is more historically accurate than *History*, because *Tom a Lincoln* had not been written when Geoffrey of Monmouth was writing his work.

C. Geoffrey of Monmouth's accounts of Arthurian legends in *History* are more similar overall in content to the accounts in *Tom a Lincoln* than they are to the accounts in *Le Morte d'Arthur*.

D. Malory encountered the Round Table in a source that Geoffrey of Monmouth was not familiar with when writing his *History*.

Information & idea

Command of Evidence

Ans D

15. Prolonged exposure to anthropogenic noise (sounds from human sources like traffic or mining) can affect animals, as Kirsty Elizabeth McLaughlin and Hansjoerg P. Kunc found in a 2015 study of zebra cichlids. Researchers conducted a meta-analysis of studies of how such noise affects animals and found that, for every study, relevant traits or behaviors of the animals were observably different between the exposed group and the otherwise similar but unexposed group. Although, on average, studies of fish showed larger differences than studies of birds did, for every class of animals examined, there were individual studies showing differences well above the average for fish. Therefore, the results of the meta-analysis suggest that _____

Which choice most logically completes the text?

A. the studies in the meta-analysis that examined fish were more likely than those that examined birds to specify whether the observed effects were detrimental.

B. the difference found in the study conducted by Kirsty Elizabeth McLaughlin and Hansjoerg P. Kunc was likely larger than the average difference for studies of zebra cichlids included in the meta-analysis.

C. some studies of birds found larger effects of exposure to anthropogenic noise than some studies of fish did.

D. the differences that studies attribute to exposure to anthropogenic noise are likely to be more pronounced for birds than they are for fish.

Information & idea

Command of Evidence

Ans C

16.The term *retroflex* derives from Latin and means “bent back,” an apt descriptor for the branch of consonants—retroflex consonants—pronounced with the tongue curling up and back in the mouth. In many languages, including English, these consonants are ____; in some dialects of Mandarin, however, four such consonants (“ch,” “sh,” “zh,” and “r”) are relatively common.

Which choice completes the text so that it conforms to the conventions of Standard English?

A. rare,

B. rare

C. rare:

D. rare and

Standard English Convention

Boundaries

Ans B

17.Along with carbon dioxide concentration and temperature, light intensity affects the chemical reaction rate of ____; as light intensity increases, so does the rate at which the reactants (water and carbon dioxide) are converted into their products (glucose and oxygen).

Which choice completes the text so that it conforms to the conventions of Standard English?

A. photosynthesis

- B. photosynthesis:
- C. photosynthesis and
- D. photosynthesis,

Standard English Convention

Boundaries

Ans A

18. The fungus *Scleroderma citrinum*, commonly referred to as common earthball, is known to establish ectomycorrhizal bonds with certain oak tree species; in many cases, such symbiotic relationships, wherein fungal hyphae make contact with the oak's root system, _____ extensive networks, which can be used to transport nutrients between trees.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A. to form
- B. having formed
- C. form
- D. forming

Standard English Convention

Parallalism

Ans C

19. Though he would go on to work with many respected musicians, including Paul Hanson on his album *Frolic in the Land of Plenty*, American _____ may be best known for his time with Sugar Hill Records, where he helped craft the sound of early hip-hop music.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A. drummer, Dennis Chambers
- B. drummer Dennis Chambers,
- C. drummer Dennis Chambers

D. drummer, Dennis Chambers,

Standard English Convention

Boundaries

Ans C

20. Stephen Graham Jones's short essay on "hueco," a word referring to a water-retaining depression in a rock face, is included in the collection *Home Ground: A Guide to the American Landscape*. The book wasn't written solely by _____ other authors, such as Pamela Frierson and Barry Lopez, also contributed essays.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A. Jones, however;
- B. Jones, however,
- C. Jones; however,
- D. Jones. However,

Standard English Convention

Boundaries

Ans C

21. The Organisation for Economic Co-operation and Development (OECD) tracks comparative price list data for its thirty-eight member countries. According to this data, in July 2021, a "basket" of goods and services priced at 100 US dollars (USD) in the United States would have cost 72 USD in fellow OECD member _____

Which choice completes the text so that it conforms to the conventions of Standard English?

- A. nation, Portugal.
- B. nation—Portugal.
- C. nation; Portugal.
- D. nation Portugal.

Standard English Convention

Boundaries

Ans A

22. In his painting *At the Cycle-Race Track*, Jean Metzinger aims to depict a bike race in four-dimensional space. Of course, Metzinger's painting doesn't technically represent a fourth dimension; humans can only see in three dimensions. _____ by depicting the race through multiple, simultaneous perspectives, Metzinger offers a fascinating glimpse at what this other universe might look like.

Which choice completes the text with the most logical transition?

- A. For example,
- B. That said,
- C. In other words,
- D. Moreover,

Expression of Idea

Transition

Ans B

23. In 2015, geologist Peter Rogerson recalculated the geographic center of Colorado, which had previously been determined in 1920 by the US Geological Survey (USGS). The USGS, Rogerson contended, had omitted important considerations, not least the curvature of the earth. _____ the state's USGS-determined center—a point 40 miles west of the municipality of Colorado Springs—was inaccurate.

Which choice completes the text with the most logical transition?

- A. Hence,
- B. Next,
- C. That being said,
- D. However,

Expression of Idea

Transition

Ans A

24. On March 8, 1971, the Apollo 14 crew received a ticker-tape parade in New York City in recognition of their mission to the Moon. Of the 206 ticker-tape parades held between 1886 and 2022, a number were for the US space program. _____ the Apollo 14 parade was just one of 10 honoring astronauts.

Which choice completes the text with the most logical transition?

- A. Therefore,

B. Nevertheless,

C. In addition,

D. Indeed,

Expression of Idea

Transition

Ans D

25. While researching a topic, a student has taken the following notes:

- The Bauhaus was an influential German art and design school founded in 1919 by Walter Gropius.
- Gropius was the school's director from 1919 to 1928.
- Hannes Meyer was its director from 1928 to 1930.
- Ludwig Mies van der Rohe was its director from 1930 to 1933.
- Stefan Wolpe was a composer who began attending the Bauhaus in 1923.
- Iwao Yamawaki was a photographer and architect who began attending the Bauhaus in 1930.

The student wants to emphasize the order in which the two students attended the Bauhaus. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. Walter Gropius founded the Bauhaus in 1919 and acted as its director until 1928, at which point Hannes Meyer succeeded him in the role.

B. Stefan Wolpe and Iwao Yamawaki were both Bauhaus students, but Wolpe, who began attending in 1923, preceded Yamawaki.

C. Founded in 1919, the Bauhaus was a German art and design school whose past students included Stefan Wolpe and Iwao Yamawaki.

D. The Bauhaus was founded in 1919, and Iwao Yamawaki began attending in 1930.

Expression of Idea

Rhetorical Synthesis

Ans B

26. While researching a topic, a student has taken the following notes:

- Generally, an object will heat up when twisted.
- The twisting of an object is known as torsion.

- A 2019 study led by Zunfeng Liu and Ray Baughman tested the torsional heating of various fibers.
- When a 1-millimeter-thick sample of dimethicone fiber was twisted, its average surface temperature increased by 1.8°C.
- When a 4-millimeter-thick sample of styrene-ethylene-butylene-styrene (SEBS) rubber fiber was twisted, its average surface temperature increased by 3.5°C.

The student wants to contrast the two samples. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A. The SEBS rubber sample used in the 2019 study was thicker than the dimethicone sample.
- B. Twisting an object will generally cause its temperature to increase, a process known as torsional heating.
- C. When the fibers were twisted as part of the 2019 study, the surface temperature of both samples increased.
- D. In 2019, researchers studied the effect of torsional heating on various fibers, including samples of SEBS rubber and dimethicone.

Expression of Idea

Rhetorical Synthesis

Ans A

27. While researching a topic, a student has taken the following notes:

- Element abundance is a relative measure of the amount of a chemical element in a given environment.
- Abundance can be expressed as a mass fraction.
- A mass fraction is the ratio of the total mass of an element to the combined mass of all elements in a given environment.
- The mass fraction of iron (Fe) on Earth is 319,000 parts per million (ppm).
- The mass fraction of strontium (Sr) on Earth is 13 ppm.

The student wants to specify the total mass of Earth's iron relative to that of all elements. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A. The mass fraction of iron on Earth is 319,000 ppm.
- B. Iron's mass relative to the combined mass of all elements on Earth is 13 ppm.
- C. Iron's abundance can be expressed by a mass fraction, a measure of the element's mass relative to that of all elements on Earth.
- D. Iron's abundance on Earth is greater than strontium's, as reflected by iron's higher mass fraction.

Expression of Idea

Rhetorical Synthesis

Ans A

Module 03

1. Cybersecurity experts often encourage users to create passwords that are fairly complicated and therefore difficult to guess. Nonetheless, research has shown that the more _____ approach to password selection seems to favor convenience over security: for example, the eighth most commonly used password in 2019 was the easily remembered "iloveyou."

Which choice completes the text with the most logical and precise word or phrase?

- A. Prevailing
- B. Byzantine
- C. Inventive
- D. Intricate

Ans A

Craft & structure

Words in context

2. The fossil remains of the individual known as Denisova 8, discovered in Russia in 2010, can help paleoanthropologists not only _____ steps in the evolution of hominids but also illuminate the Pleistocene epoch generally, revealing important details about the time in which Denisova 8 lived.

- A. beget
- B. commandeered
- C. privilege
- D. discern

Ans D

Craft & structure

Words in context

3. As exemplified by Sakha songs about local ecosystems and Kaluli songs about rainforest sounds, ecological information can be transmitted in Indigenous songs, and in some instances is maintained only in this way. Kwaxsistalla Wathl'thla, a song keeper for the Kivakwaka'wakw people in Canada, collaborated with ethnobiologist Dana Lepofsky et al., sharing songs referencing terraced intertidal clam gardens the people implemented in the past to foster healthy development of a dietary staple. Drawing on archaeological evidence as well, Lepofsky et al., determined that the prevalence of the practice described in the songs corresponded with growth in clam size and abundance despite increased harvesting pressure- a finding that demonstrates that _____

Which choice most logically completes the text?

A the clams harvested from intertidal terraces by Kwakwaka'wakw people in the past likely were a different species than the clams found in those areas today.

B the Kwakwaka'wakw people likely would not have detailed their creation of clam gardens in songs if their efforts had not produced significantly larger clams.

C representation of practical applications of ecological knowledge is the defining characteristic of the music of certain Indigenous peoples.

D effective methods for the cultivation of sources of sustenance are among the ecological knowledge preserved in indigenous songs.

Ans D

Information & Idea

Inference

4. The Clarion-Clipperton Zone (CCZ) is an expanse of abyssal plain and seamounts (underwater mountains) between Hawai'i and Mexico in which mining is permitted, but the area's biodiversity is poorly understood. The vast majority of invertebrate species found in a recent survey of the CCZ were hitherto unknown to scientists, and sampling for animal life has been highly concentrated in the eastern part of the zone. Some species like *Ledella knudseni* were identified both morphologically and molecularly (using DNA analysis), whereas *Prionospio branchilucida* and other species for which molecular sampling was impracticable were identified solely based on physical morphology. Since molecular analysis does not require distinguishing subtle physiological differences, it is _____

Q. Which choice most logically completes the text?

A more likely to properly distinguish between morphologically similar but distinct invertebrate species than is the method used to identify *Prionospio branchilucida*.

B more frequently used for identifying species with physiological characteristics like those of *Ledella knudseni* than morphological analysis is.

C better suited to distinguishing *Prionospio branchilucida* from morphologically similar species than to distinguishing *Ledella knudseni* from morphologically similar species. *Prionospio branchilucida*.

D less likely to differentiate among morphologically similar invertebrate species than traditional morphological identification is, because DNA-based methods depend on detecting subtle physical traits that may be hard to observe in preserved specimens.

Ans A

Information & Idea

Inference

5. Scholars cite *One Hundred Years of Solitude*, the 1967 novel by Colombian author Gabriel García Márquez, as a foundational text of magical realism, the Latin American style of fiction in which antirealistic plot-devices—often borrowed from the folkloric traditions of Indigenous and colonial societies in the Americas—are deployed in an otherwise realistic mode of representation typical of the modern novel. This style has exerted a decisive influence on authors around the world, including Olga Tokarczuk, whose 1996 novel *Primeval and Other Times resembles classic magical realist novels in its juxtaposition of literary realism with folklore namely, that of Poland.*

Q. Which quotation from a literary scholar would most directly support the claim in the underlined portion of the text?

A "Like many works in the Latin American magical realist tradition, *Primeval and Other Times* is indebted to antirealistic elements in the folkloric tradition of Poland."

B "The logic of the realistic plot of *Primeval and Other Times* is repeatedly and productively disrupted by the presence of imagery and situations drawn from Polish folklore."

C "While *Primeval and Other Times* alternates between realistic and antirealistic modes of representation, details suggesting the influence of Polish folklore nevertheless occur throughout the novel."

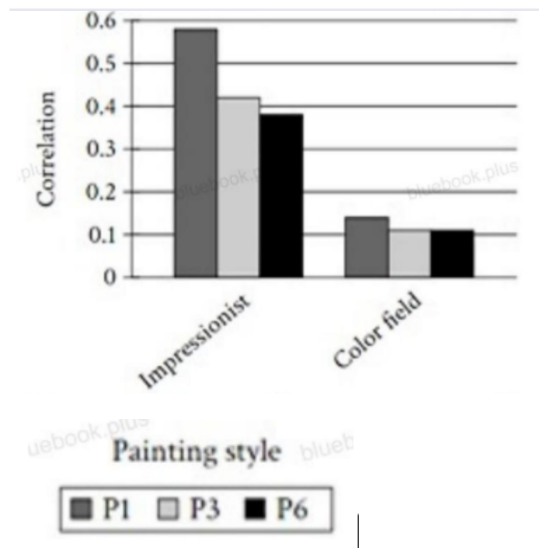
D "Although Polish folklore clearly informs the style and occasionally antirealistic plot of *Primeval and Other Times*, the novel also shows the inarguable influence of the magical realist tradition of Latin America."

Ans A

Craft & structure

Text Structure & Purpose

6. Correlation between Model-Predicted and Participant-Reported Enjoyment Ratings, by Painting Style



Neuroscientist Kiyohito Ligaya and colleagues developed a computational model to predict how much a person will enjoy a particular work of art on a scale from 1 (not at all) to 4 (very much). They then recruited participants to use the same scale to rate several sets of paintings in various styles and calculated the correlation between the ratings predicted by the model and those reported by the participants. Assuming participant P1 gave ratings to the impressionist and color-field paintings, the data in the graph suggest that the model predicted that _____

Q.Which choice most effectively uses data from the graph to complete the statement?

- A P1's and P2's ratings for impressionist paintings would equal one another.
- B P1's and P2's ratings for impressionist paintings would differ from one another.
- C P1 would derive more aesthetic pleasure from impressionist paintings than from cubist paintings.
- D P1 would derive less aesthetic pleasure from impressionist paintings than from cubist paintings.

Ans B

Information & Idea

Command of Evidence

7.Honda's introduction of a walk-behind lawnmower in 1978 is an instance of brand extension-the company leveraged its brand recognition as an automobile manufacturer to enter a product category where it had not previously competed. To determine if perceived category similarity predicts consumers' likelihood of purchasing brand extensions, Alicia Grasby et al. identified 30 extended-brand pairs

(e.g., the same brand of hand lotion and mouthwash) in 52 weeks of purchases by approximately 60,000 households; for each pair, Grasby et al. had consumers rate the similarity of the product categories and calculated the change in probability of a brand in one category being purchased if the same brand was purchased in the other category.

Q. Which finding, if true, would provide evidence that the possibility Grasby et al. investigated does occur?

A Consumers' ratings and the changes in probability calculated by the researchers were both affected by level of brand recognition.

B There was a strong positive correlation between consumers' ratings and the changes in probability calculated by the researchers.

C Consumers' ratings varied substantially by category pair, whereas the changes in probability calculated by the researchers were broadly similar for each pair.

D Consumers tended to purchase more products in categories in which extended-brand pairs are found than in categories in which extended-brand pairs are not found.

Ans B

Information & Idea

Command of Evidence

8. The utilization of deceptive antipredator displays is well documented in the little ringed plover (*Charadrius dubius*) and other species of the avian order Charadriiformes. An extensive literature review conducted by Léna de Framond et al. revealed that this trait has evolved across a surprisingly large phylogenetic distribution of 13 Aves orders, including Caprimulgiformes and Gruiformes. Subsequent investigation of potential selection mechanisms prompted the researchers to conclude that independent of avian order, the prevalence of the trait is mediated by environmental variations associated with the absolute latitude of brooding sites.

Q. Which finding, if true, would most strongly support the researchers' conclusion?

A. Across the orders in the study, approximately 54% of the bird species brood in ranges from 50° to 80° absolute latitude, but most of the birds that are known to use deceptive antipredator displays brood between 0° and 30° absolute latitude.

B. Deceptive antipredator displays are documented in Charadriiformes species across the entire range of absolute latitudes of brooding sites within that order, but in species from other orders, deceptive antipredator displays are documented only

when brooding sites are at absolute latitudes 10°—20° higher than what is typical for those species.

C. The use of deceptive antipredator displays is widespread among Charadriiformes species independent of the absolute latitude of their breeding sites, but its prevalence in other avian orders is limited to species with brooding sites located in absolute latitudes of 0°–30°.

D. Across the orders in the study, deceptive antipredator displays are observed in approximately 34% of species with brooding ranges of 0°–30° absolute latitude and approximately 60% of species with brooding ranges of 50°–80° absolute latitude.

Ans D

Information & Idea

Command of Evidence

9. Nicolas Dussex and colleagues relied on historical DNA (hDNA) genomic data incidentally preserved in specimens housed in natural history collections to investigate the evolutionary trajectory of the kakapo parrot (*Strigops habroptilus*). Although this approach offers unique benefits, such as the ability to sample genomic material from extinct species, it remains relatively underutilized because archival specimens are sometimes stored in ways that compromise DNA quality, a situation not easily remediable under current methodological paradigms and with extant DNA extraction and analysis technologies.

Q. What does the text most strongly suggest about specimens from natural history collections?

A While they were used in a study that made an important scientific discovery, they are generally of marginal value as sources of genomic data.

B Because of their often-deteriorated condition, they tend to yield genomic data that are time-consuming to extract and interpret.

C They are primarily used as sources of genomic data by scientists studying organisms that have undergone major evolutionary change.

D While they may contain valuable genomic data, not all of them can yield usable DNA.

Ans D

Information & Idea

Detail & Central Idea

In parts of Australia, the feral cat is a major predator of the ash-grey mouse. Researcher Danielle A. Risbey and colleagues found that when this predation pressure on ash-grey mice was temporarily reduced, their numbers significantly increased. This finding illustrates a foundational ecological principle: predators control prey population numbers.

Text 2

Michel Crête and Hélène Jolicoeur found that excluding wolves from a site in Quebec, Canada, where they typically prey on moose had no significant effect on moose abundance. Many other predation relief studies show an increase in prey abundance, but those studies often focus on small, rapidly reproducing prey, like birds, mice, and rabbits, rather than large, slowly reproducing prey, like moose, which could account for the difference between those results and Crête and Jolicoeur's results.

Q. Based on the texts, the author of Text 2 would most likely agree with which statement about Risbey and colleagues' findings (Text 1)?

A They may be partially attributable to physical characteristics of ash-grey mice that are not shared by moose.

B They may have been different if the ratio of the body size of feral cats to that of ash-grey mice was greater than it is.

C They may be partially explained by the fact that feral cats are smaller and reproduce more rapidly than wolves.

D They may reflect a response that does not tend to be observed in prey species other than ash-grey mice.

Ans A

Craft & structure

Cross text connection

10. Zurich has high pedestrian traffic, but simply replicating a feature of Zurich associated with walkability -eg., its high number of street crossings-may be insufficient to induce increased walking in other cities. As urbanist Mariela Alfonzo argues, our understanding of individuals decision-making about whether to walk is insufficiently robust: some studies emphasize the role of demographic characteristics, others the role of neighborhood type, and so on, but walking decisions are made in complex contexts in which multiple conditions and needs inform individuals' choices.

Q.Which choice best describes the function of the references to "demographic characteristics" and "neighborhood type" in the text as a whole?

A They illustrate factors that researchers believe people consider when making walking decisions in most contexts but that the text argues are unique to walking decisions made by people in Zurich.

B They are examples of factors that studies suggest are important in people's decision-making about walking but that the text claims most people rarely consider when making walking decisions.

C They represent factors that have been identified as important influences on walking decisions but that the text suggests are merely some of the many factors that may contribute to people's decision-making about walking.

D They identify factors that Alfonzo argues have been overemphasized in studies of decisionmaking about walking but that the text asserts are relevant to most people's walking decisions.

Ans C

Craft & structure

Text Structure & Purpose

11. Mauricio Drelichman and Hans-Joachim Voth's analysis of the overall debt and revenue of the government of Philip II (who ruled an empire including Spain and Sicily from 1556 to1598) found an intriguing ____ : although the government regularly defaulted on debt, it ran an even larger overall surplus than did the government of eighteenth-century Britain,which historians consider a model of fiscal virtue.

Which choice completes the text with the most logical and precise word or phrase?

A harbinger

B exemplar

C sanction

D incongruity

Ans D

Craft & structure

Words in context

12. The Organisation for Economic Co-operation and Development (OECD) tracks comparative price list data for its thirty-eight member countries. According to this data, in July 2021, a "basket" of goods and services priced at 100 US dollars (USD) in the United State would have cost 80 USD in fellow OECD member _____

Which choice completes the text so that it conforms to the conventions of Standard English?

A nation Spain.

B nation, Spain.

C nation-Spain.

D nation: Spain.

Ans A

Standard Eng Convention

Punctauation

13. The Suzuki method is an approach to music instruction _____ to teach music in the same natural way children learn language, emphasizes retaining and reviewing previously learned pieces of music. In Suzuki piano instruction, students begin their musical education with simple folk songs like "The Honeybee."

Which choice completes the text so that it conforms to the conventions of Standard English?

A that, is aiming

B that, aiming

C that aims

D that has aimed

Ans B

Standard Eng Convention

Punctauation

14.Many works of the Roman historian Suetonius (1st century CE), including his book on miscellaneous topics, Pratum, have been lost to time. Antiquarians _____ to read one of Suetonius's extant works-not one known only through references to it by other authors, that is -might instead procure a copy of his De via caesarum, a biography of the Roman Empire's first leaders.

Q.Which choice completes the text so that it conforms to the conventions of Standard English?

- A hope
- B hoped
- C have hoped
- D hoping

Ans D

Standard English Convention

Verbs

15. Charline Effah is a novelist from Minvoul, Gabon, who has earned international acclaim in recent years. Writers of other literary forms, such as poetry and drama, are less likely to be recognized beyond their nations' borders, but many are still well _____ Malika Ouattara, for example, is a celebrated slam poet from Burkina Faso.

Q.Which choice completes the text so that it conforms to the conventions of Standard English?

- A known
- B known,
- C known:
- D known and

Ans C

Standard Eng Convention

Punctaution

16.Maintaining road infrastructure in the United States is demanding because of the country's extensive system of _____ traffic patterns can be unpredictable in major metropolitan areas.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A highways and the fact that,

B highways, and the fact that,
C highways, and, the fact that
D highways and the fact that

Ans D

Standard Eng Convention

Punctauation

17. Emily Carter, program coordinator for the Austin Public Library in Texas, helped launch the library's first targeted literacy initiative, Read _____ Texan Tales, in 2014; and a summer reading symposium in 2018.

Q.Which choice completes the text so that it conforms to the conventions of Standard English?

- A Now; in 2011, a digital archive;
- B Now; in 2011, a digital archive,
- C Now, in 2011; a digital archive,
- D Now, in 2011, a digital archive,

Ans A

Standard Eng Convention

Punctauation

18. Approximate Operational Frequency and Throughput for Five Processor Architectures

Architect ure	Operational frequency (GHz)	Processing throughput (BIPS)
Model C	3.8	16.5

Model B	4.2	18.0
Model A	2.5	18.2
Model D	2.1	14.1
Model E	2.9	14.8

A team of computer engineers conducted a study to determine how fast different processor architectures typically operate (in gigahertz, or GHz) and how much data they can effectively process (in billion instructions per second, or BIPS). They found that, although architectures vary widely in their operational frequency, the effective processing throughput tends to vary much less. Thus, they claim that two architectures with very different operational frequencies can nonetheless achieve the same processing throughput.

Which choice best describes data from the table that support the engineers' claim?

A Among the five architectures in the table, Model D and Model E have the lowest operational frequencies and the lowest processing throughput.

B Model A achieves a processing throughput approximately the same as Model B despite operating at a lower frequency.

C Among the five architectures in the table, the architecture that operates at the highest frequency is also the architecture that achieves the highest throughput.

D Model C and Model B operate at approximately the same frequency, but Model C processes data faster than Model B does.

Information & Idea

Command of Evidence

Ans A

19.The ruins of Mesa Verde in Colorado, which were inhabited by Ancestral Puebloans centuries ago, remain a focal point for historians. Regrettably, as _____point out, environmental weathering has worn down the sandstone masonry, causing significant architectural details to disappear.

Q.Which choice completes the text so that it conforms to the conventions of Standard English?

A preservationists, David Wright and Emily Carter,

- B preservationists, David Wright and Emily Carter
- C preservationists David Wright and Emily Carter
- D preservationists David Wright, and Emily Carter

20. In analyzing the architectural designs of American architect Frank Lloyd Wright, _____ have overlooked his equally significant contributions to urban planning theories.

Q. Which choice completes the text so that it conforms to the conventions of Standard English?

- A many scholars have focused on Wright's use of organic style but
- B Wright's use of organic style has been the focus of many scholars, who
- C there are many scholars who have focused on Wright's use of organic style, but they
- D the focus of many scholars has been on Wright's use of organic style; they

21. When, in the early 1900s, volcanologists first speculated that a particular supervolcano erupted in a cyclical pattern, the idea was met with skepticism. Each massive eruption is so devastating, though, that it typically destroys the geological strata deposited by the eruption before it. _____ scientists were unable to validate the hypothesis of cyclical eruptions until deep-earth scanning technology was invented.

Which choice completes the text with the most logical transition?

- A Hence,
- B Moreover,
- C Nevertheless,
- D Next,

22. While researching a topic, a student has taken the following notes:

- When the Statue of Liberty was dedicated in New York in 1886, it was a deep brown color.
- Journalists at the time compared its appearance to that of a new penny.
- Today, the statue is famous for its distinctive green hue.
- Oxidation caused a green layer of carbonate minerals, called a patina, to form on the copper surface.

The student wants to specify the reason the Statue of Liberty's color changed. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A When the Statue of Liberty was dedicated in 1886, journalists compared its deep brown color to the appearance of a new penny.

B Although it was a deep brown color when dedicated in 1886, the Statue of Liberty appears distinctively green today.

C In the time since the Statue of Liberty appeared brown in 1886, oxidation has caused a green patina to form on the copper surface.

D The Statue of Liberty is made of copper, but oxidation has created a green patina that covers the brown surface.

23. While researching a topic, a student has taken the following notes:

- Cetaceans, which include whales, dolphins, and porpoises, live exclusively in water.
- Cetaceans are descended not from aquatic animals but from four-legged, land-dwelling mammals.
- American paleontologist Dr. Sarah Jenkins recently recovered a fossil with four legs, hooved feet, and the distinct ear bones of a whale.
- Jenkins refers to her unique discovery as a "transitional specimen."
- The fossil exemplifies a pivotal period in the evolution of cetaceans from their terrestrial predecessors.

The student wants to emphasize the fossil's significance. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A American paleontologist Dr. Sarah Jenkins's fossil has the ear bones of a whale, which, like dolphins and porpoises, is a cetacean.

B Cetaceans are descended from four-legged, land-dwelling mammals; a fossil that resembles both groups was recently found.

C Possessing four legs but the ear bones of a whale, the unique fossil exemplifies a pivotal period in the evolution of cetaceans from their terrestrial predecessors.

D A "transitional specimen" was recently found by paleontologist Dr. Sarah Jenkins.

24 While researching a topic, a student has taken the following notes:

- Arthur Evans is an artist.
- He specializes in illustrations of historic American bridges.
- An illustration is an artistic representation of a subject.
- Evans completed a woodcut of the Brooklyn Bridge in 1988.

- He completed an acrylic painting of the Golden Gate Bridge in 2002.

The student wants to emphasize a difference between the two illustrations. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A The illustrations, or artistic representations, of the Brooklyn Bridge and the Golden Gate Bridge were both completed by Evans in the late 20th and early 21st centuries.

B Evans has completed illustrations of numerous historic bridges, including the Brooklyn Bridge and the Golden Gate Bridge.

C While both are by Evans, the 1988 illustration of the Brooklyn Bridge is a woodcut, and the 2002 illustration of the Golden Gate Bridge is an acrylic painting.

D The Brooklyn Bridge is located in New York City, while the Golden Gate Bridge spans the strait connecting San Francisco Bay and the Pacific Ocean.

Expression of idea

Rhetorical Synthesis

Ans C

25. While researching a topic, a student has taken the following notes:

- In the 1850s, a British naval officer created a manuscript chart of the Alaskan coast.
- He wrote "? Name" next to an unnamed cape to query its proper identity.
- A mapmaker later misinterpreted this annotation as "C. Nome" (Cape Nome).
- The name stuck and was eventually applied to the nearby city of Nome, Alaska.
- The name originated solely from this handwriting error.

Which choice most effectively uses relevant information from the notes to accomplish this goal?

A A British naval officer wrote "? Name" on a chart to verify the identity of an unnamed cape in Alaska.

B The city of Nome is located near a cape on the Alaskan coast that was charted in the 1850s.

C In the 1850s, a mapmaker created a nautical chart of the Alaskan coast and labeled a specific point "C. Nome."

D Mistaking the annotation "? Name" for a specific label, a mapmaker accidentally assigned the name Nome to an Alaskan cape.

Expression of idea

Rhetorical Synthesis

Ans D

26. While researching a topic, a student has taken the following notes:

- In 1801, French artist Jacques-Louis David painted *Napoleon Crossing the Alps*.
- His large painting (approximately 102×87 inches) depicts the French leader on horseback crossing a mountain pass.
- In 2005, American artist Kehinde Wiley painted *Napoleon Leading the Army over the Alps*.
- Wiley's large painting (108×108 inches) was inspired by David's work.
- It portrays a young Black man in contemporary streetwear riding a rearing horse.

The student wants to emphasize a similarity between the two paintings. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A Wiley, an American artist, finished his painting in 2005; David, a French artist, completed his in 1801.

B Although Wiley's painting was inspired by David's, the attire of the figures in the two works is very different.

C David's and Wiley's paintings are both large, measuring approximately 102×87 inches and 108×108 inches, respectively.

D David's painting depicts a 19th-century military leader, while Wiley's depicts a contemporary rider in streetwear.

Expression of idea

Rhetorical Synthesis

Ans C

27. While researching a topic, a student has taken the following notes:

- In 1931, American artist Grant Wood painted *American Gothic*.
- The painting shows two figures standing in front of a farmhouse; one holds a pitchfork.
- In 1970, American artist Gordon Parks photographed *American Gothic*, *Washington, D.C.*
- Parks's photograph was inspired by Wood's painting.
- The photograph shows a Black woman standing with a mop and broom in front of an American flag.

The student wants to emphasize a similarity between Wood's painting and Parks's photograph. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) Wood created a painting in 1931, whereas Parks created a photograph in 1970.
- B) Parks's work was inspired by Wood's, but it replaces a farmhouse setting with an American flag.
- C) Both works depict a person holding a long-handled object while standing in a posed scene.
- D) Wood's work portrays rural life, while Parks's portrays a worker in Washington, D.C.

Expression of idea

Rhetorical Synthesis

Ans C

Module 1:

Question 1

A courier company charges a fixed delivery fee plus a constant cost per mile. A delivery that is 30 miles long costs \$42 more than a delivery that is 18 miles long. The 30-mile delivery costs 1.4 times as much as the 18-mile delivery.

What is the cost, in dollars, of a 44-mile delivery?

- A. 189
- B. 196
- C. 203
- D. 210

Answer: B

Domain: Algebra

Subdomain: Linear functions

Difficulty: Difficult

Question 2

A theater sells adult tickets and student tickets. The number of student tickets sold was 80 more than the number of adult tickets sold. Adult tickets cost \$18 each, and student tickets cost \$10 each. If the total ticket revenue was \$4,440, how many tickets were sold in total?

- A. 300
- B. 320
- C. 340
- D. 360

Answer: C

Domain: Algebra

Subdomain: Systems of 2 linear equations in 2 variables

Difficulty: Medium

Question 3

A repair company charges a fixed inspection fee of \$25 plus \$18 per hour of work. A customer has a budget of at most \$160 for the repair.

What is the greatest whole number of hours of work the customer can afford?

- A. 6
- B. 7
- C. 8
- D. 9

Answer: B

Domain: Algebra

Subdomain: Linear inequalities in 1 or 2 variables

Difficulty: Easy

Question 4

A storage facility uses small crates and large crates. Each small crate provides 4 cubic feet of storage and costs \$35. Each large crate provides 9 cubic feet of storage and costs \$70. The facility needs at least 120 cubic feet of storage, can spend at most \$980, must use at least 8 large crates, and can use at most 10 small crates.

Which ordered pair (s, l) could represent the number of small crates and large crates used?

- A. (8, 9)
- B. (10, 9)
- C. (12, 8)
- D. (14, 7)

Answer: B

Domain: Algebra

Subdomain: Linear inequalities in 1 or 2 variables

Difficulty: Difficult

Question 5

A company's monthly profit P , in dollars, from selling a product at price p dollars is modeled by

$$P = -4p^2 + 240p - 2,000$$

According to the model, what price p gives the maximum monthly profit?

- A. 24
- B. 28
- C. 30
- D. 36

Answer: C
Domain: Advanced Math
Subdomain: Nonlinear functions
Difficulty: Easy

Question 6

Two positive numbers have a sum of 14. The sum of their squares is 116. What is the product of the two numbers?

- A. 36
- B. 40
- C. 44
- D. 48

Answer: B
Domain: Advanced Math
Subdomain: Systems of equations in 2 variables
Difficulty: Easy

Question 7

The concentration $C(t)$, in milligrams per liter, of a medicine t hours after it is taken is modeled by

$$C(t) = 80/(t + 4)$$

A second version of the medicine delays the concentration pattern by 3 hours and has concentration 5 milligrams per liter less than the original model after the delay. This second concentration is modeled by $D(t) = C(t - 3) - 5$, for $t > 3$.

What is $D(7)$?

- A. 3
- B. 5
- C. 7
- D. 10

Answer: B
Domain: Advanced Math
Subdomain: Nonlinear functions
Difficulty: Medium

Question 8

A bacterial culture is modeled by an exponential function $N(t) = ab^t$, where t is measured in hours. The culture has 2,700 bacteria after 6 hours and 218,700 bacteria after 18 hours.

How many bacteria were in the culture initially?

- A. 150
- B. 300
- C. 450
- D. 900

Answer: B

Domain: Advanced Math

Subdomain: Nonlinear functions

Difficulty: Medium

Question 9

The height h , in feet, of a launched object t seconds after launch is modeled by a quadratic function. The object would reach ground level at $t = -2$ and $t = 14$, and its maximum height is 512 feet.

According to the model, what was the height of the object at the time it was launched?

- A. 112
- B. 168
- C. 224
- D. 256

Answer: C

Domain: Advanced Math

Subdomain: Nonlinear functions

Difficulty: Medium

Question 10

A square display screen has side length $\sqrt{s + 84}$ inches. If the side length is equal to $s - 6$ inches, where s is positive, what is the value of s ?

- A. 10
- B. 12
- C. 15
- D. 18

Answer: D

Domain: Advanced Math

Subdomain: Nonlinear equations in 1 variable

Difficulty: Easy

Question 11

The table below summarizes students by grade level and club membership.

Grade level	Debate club	Robotics club	Total
9th or 10th grade	18	x	$x + 18$
11th or 12th grade	32	24	56
Total	50	$x + 24$	$x + 74$

If one student is selected at random, the probability that the student is in 11th or 12th grade, given that the student is in the robotics club, is $\frac{3}{5}$. What is the value of x ?

- A. 12
- B. 16
- C. 20
- D. 24

Answer: B

Domain: Problem-Solving and Data Analysis

Subdomain: Probability and conditional probability

Difficulty: Medium

Question 12

A data set has mean 62 and standard deviation 8. A new data set is created by replacing each value x with $1.5x + 12$.

What are the mean and standard deviation of the new data set?

- A. Mean = 93, standard deviation = 8
- B. Mean = 93, standard deviation = 12
- C. Mean = 105, standard deviation = 8
- D. Mean = 105, standard deviation = 12

Answer: D

Domain: Problem-Solving and Data Analysis

Subdomain: One-variable data: distributions and measures of center and spread

Difficulty: Medium

Question 13

A data set consists of 9 distinct positive integers. The mean is 28, the median is 25, and exactly three values in the data set are greater than 40.

What is the greatest possible value of the largest integer in the data set?

- A. 96
- B. 102
- C. 108
- D. 114

Answer: C

Domain: Problem-Solving and Data Analysis

Subdomain: One-variable data: distributions and measures of center and spread

Difficulty: Difficult

Question 14

A table summarizes a data set of 30 integer values using the intervals shown. Each interval includes its lower endpoint and excludes its upper endpoint.

Interval	Frequency
0–9	3
10–19	5
20–29	9
30–39	7
40–49	6

What is the greatest possible mean of the data set?

- A. 31.25
- B. 31.67

- C. 32.10
- D. 32.50

Answer: B

Domain: Problem-Solving and Data Analysis

Subdomain: One-variable data: distributions and measures of center and spread

Difficulty: Difficult

Question 15

For a particular equilateral triangle, the numerical value of its area is equal to twice the numerical value of its perimeter. What is the perimeter of the triangle?

- (A) $12\sqrt{3}$
- (B) $18\sqrt{3}$
- (C) $24\sqrt{3}$
- (D) $36\sqrt{3}$

Answer: D

Domain: Geometry and Trigonometry

Subdomain: Lines Angles and triangles

Difficulty: Medium

Question 16

Two similar solid figures have surface areas in the ratio 36:121. The volume of the larger solid exceeds the volume of the smaller solid by 22,300 cubic units.

What is the volume, in cubic units, of the smaller solid?

- A. 3,960
- B. 4,320
- C. 4,860
- D. 5,120

Answer: B

Domain: Geometry and Trigonometry

Subdomain: Area and volume formulas

Difficulty: Medium

Question 17

In right triangle ABC, angle C is 90° . If $\sin A = 7/25$ and $AC = 144$, what is the area of triangle ABC?

- A. 2,520
- B. 3,024
- C. 3,456
- D. 4,032

Answer: B

Domain: Geometry and Trigonometry

Subdomain: Right triangles and trigonometry

Difficulty: Medium

Question 18

In the xy-plane, a circular fountain has equation

$$(x - 5)^2 + (y + 2)^2 = 169$$

A vertical walkway is represented by the line $x = -7$. The walkway intersects the fountain at two points. What is the greater y-coordinate of these two points?

- A. -7
- B. -3
- C. 3
- D. 5

Answer: C

Domain: Geometry and Trigonometry

Subdomain: Circles

Difficulty: Easy

Question 19

A sector of a circle has radius 15 and area 75π square units. What is the central angle of the sector, in degrees?

- A. 90
- B. 108
- C. 120
- D. 135

Answer: C

Domain: Geometry and Trigonometry

Subdomain: Circles

Difficulty: Easy

Question 20

The system of equations is shown below.

$$(x - 2)^2 + (y + 1)^2 = 50$$

$$2x - y = k$$

For how many integer values of k does the system have two distinct real solutions?

- A. 29
- B. 30
- C. 31
- D. 32

Answer: C

Domain: Advanced Math

Subdomain: Systems of equations in 2 variables

Difficulty: Elite Difficult

Question 21

The cost per member, in dollars, for a group activity is modeled by

$$f(n) = 2,400/(n + 3)$$

where n is the number of additional members who join the group. A revised model is defined by

$$g(n) = f(n + 5) + 2$$

What is the value of n where the function $y = g(n)$ is not defined?

- A. -8
- B. -5
- C. -3
- D. 2

Answer: A

Domain: Advanced Math

Subdomain: Nonlinear functions

Difficulty: Medium

Question 22

A drawer originally contains red pins and blue pins in the ratio 3:5. After 4 red pins are added and 2 blue pins are removed, the probability of selecting a red pin from the drawer is $\frac{2}{5}$.

How many pins were originally in the drawer?

- A. 96
- B. 112
- C. 128
- D. 144

Answer: C

Domain: Problem-Solving and Data Analysis

Subdomain: Ratios, rates, proportional relationships, and units

Difficulty: Medium

Module 2:

Question 1

A real estate broker earns a base salary of b dollars per year plus a 3.5% commission on the

total dollar value of properties sold, v , during that year. This year, the broker sets an explicit financial goal for their total combined annual earnings to be at least 2.5 times and at most 4 times their base salary. Which of the following inequalities represents all possible values of total property sales v , in dollars, the broker must make this year in order to achieve this goal?

- A. $1.5b \leq 0.035v \leq 3b$
- B. $(1.5b) / 0.035 \leq v \leq (3b) / 0.035$
- C. $(2.5b) / 0.035 \leq v \leq (4b) / 0.035$
- D. $2.5b \leq b + 0.035v \leq 4b$

Answer: B

Domain: Algebra

Subdomain: Linear inequalities in 1 or 2 variables

Difficulty: Medium

Question 2

For $x \neq 2.5$, the expression below is equivalent to $2x + n$, where b and n are positive integers.

$$(4x^2 + bx - 45) / (2x - 5)$$

What is the value of the product bn ?

- A. 54
- B. 72
- C. 81
- D. 90

Answer: B

Domain: Advanced Math

Subdomain: Equivalent expressions

Difficulty: Medium

Question 3

The table below summarizes the initial data points gathered from a regional voting preferences poll.

Survey Poll

Candidate	Percentage
Candidate A	25%
Candidate B	55%
Undecided	20%

A total of 5,000 randomly selected registered voters participated in this poll. In the actual election held a month later, all 5,000 of these individuals cast a ballot. Every individual who selected a specific candidate during the poll voted for that exact candidate. The individuals who were initially recorded as undecided cast their actual ballots for Candidate A in the same exact proportion as the individuals who had already committed to a specific candidate during the survey phase. To the nearest whole number, how many individuals voted for Candidate A in the actual election?

- A. 1,250
- B. 1,563
- C. 1,625
- D. 1,750

Answer: B

Domain: Problem-Solving and Data Analysis

Subdomain: Percentages

Difficulty: Difficult

Question 4

In an automated fulfillment warehouse, a high-efficiency packaging line can process a specific inventory batch in one-fifth the time it takes a manual packaging station working completely alone. Working simultaneously together, the high-efficiency line and the manual station can complete the exact same inventory batch in 4 hours. How many hours would it take the high-efficiency packaging line working completely alone to complete the batch?

- A. 0.8
- B. 4.8
- C. 5.0
- D. 24.0

Answer: B

Domain: Problem-Solving and Data Analysis

Subdomain: Ratios, rates, proportional relationships, and units

Difficulty: Difficult

Question 5

In the xy -plane, the center of circle Q is located at the coordinates $(3, -5)$. If an endpoint of a radius of circle Q is located at the point $(8, -17)$, which of the following is a valid equation for circle Q ?

- A. $(x - 3)^2 + (y + 5)^2 = 13$
- B. $(x + 3)^2 + (y - 5)^2 = 169$
- C. $(x - 3)^2 + (y + 5)^2 = 169$
- D. $(x - 8)^2 + (y + 17)^2 = 13$

Answer: C

Domain: Geometry and Trigonometry

Subdomain: Circles

Difficulty: Medium

Question 6

The value of a specialized industrial asset depreciates at a constant annual percentage rate. The function $V(m) = 14,500(0.81)^{m/24}$ gives the estimated value, in dollars, of the asset after m months of continuous operation. By what percentage does the value of the asset decrease each year?

- A. 9.0%
- B. 10.0%
- C. 19.0%
- D. 81.0%

Answer: B

Domain: Advanced Math

Subdomain: Nonlinear functions

Difficulty: Medium

Question 7

If $x^2 + y^2 - 10x + 12y + 61 = 0$, where x and y are real numbers, what is the exact value of $x - y$?

- A. -1
- B. 1
- C. 5
- D. 11

Answer: D

Domain: Geometry and Trigonometry

Subdomain: Circles

Difficulty: Medium

Question 8

In an engineering research team, the number of hardware systems engineers is exactly 25% of the rest of the technical development group. The total number of hardware systems engineers represents what percent of the entire technical development group?

- A. 20%
- B. 25%
- C. 33.33%
- D. 80%

Answer: A

Domain: Problem-Solving and Data Analysis

Subdomain: Percentages

Difficulty: Medium

Question 9

A technical drone traveled from a logistics headquarters to a remote research outpost at a constant average rate of 60 miles per hour. After completing a payload drop, the drone returned to the logistics headquarters along the exact same flight path at a constant average rate of 40 miles per hour. If the total round-trip flight time took exactly 6 hours, what was the one-way distance, in miles, from the logistics headquarters to the remote research outpost?

- A. 120
- B. 144
- C. 240
- D. 288

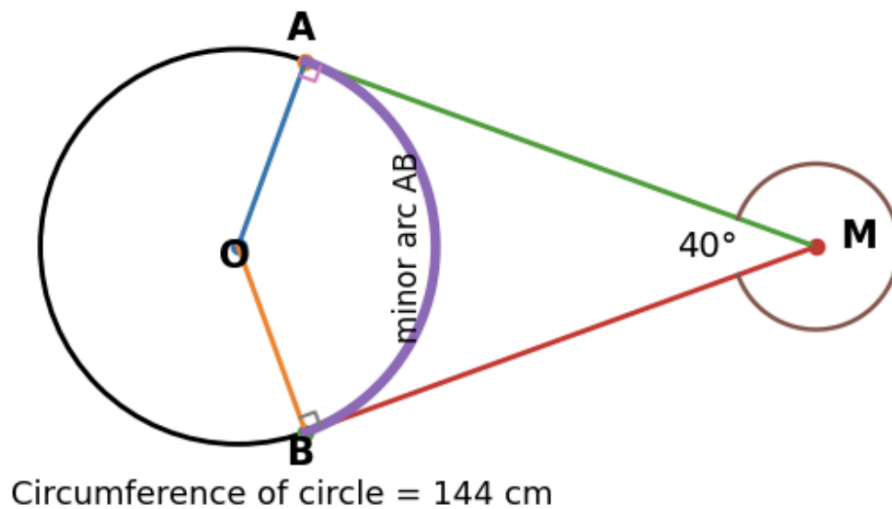
Answer: B

Domain: Problem-Solving and Data Analysis

Subdomain: Ratios, rates, proportional relationships, and units

Difficulty: Medium

Question 10



In a circle with center O, line segments MA and MB are tangent to the circle at points A and B, respectively, and intersect externally at point M. If angle AMB measures 40° and the complete circumference of the circle is 144 centimeters, what is the length, in centimeters, of major arc AB?

- A. 16
- B. 32
- C. 56
- D. 88

Answer: D

Domain: Geometry and Trigonometry

Subdomain: Circles

Difficulty: Easy

Question 11

In the xy-plane, the system of linear equations below has infinitely many solutions:

$$kx - 8y = 20$$

$$6x - ly = 30$$

where k and l are constants. What is the value of $k + l$?

- A. 4
- B. 12
- C. 14
- D. 16

Answer: D

Domain: Algebra

Subdomain: Systems of 2 linear equations in 2 variables

Difficulty: Medium

Question 12

What is the complete solution set containing all real values of x that satisfy the rational equation shown below?

$$1/(x - 3) - 2/(x + 3) = 12/(x^2 - 9)$$

- A. $\{-3\}$
- B. $\{3\}$
- C. $\{-3, 3\}$
- D. There are no real solutions.

Answer: D

Domain: Advanced Math

Subdomain: Nonlinear equations in 1 variable

Difficulty: Easy

Question 13

The partially completed contingency table below tracks the eye color distribution of 300 feline subjects in a veterinary genetics study.

Category	Gold Eyes	Green Eyes	Blue Eyes	Total
Male	$5x$	6	x	$6x + 6$
Female	$8y$	6	y	$9y + 6$
Total	246	12	42	300

Within this tracking group, there are 5 times as many males with gold eyes as males with blue eyes, and 8 times as many females with gold eyes as females with blue eyes. What percent of the total 300 subjects are either male with blue eyes or female with gold eyes?

- A. 14%
- B. 32%
- C. 42%
- D. 80%

Answer: C

Domain: Problem-Solving and Data Analysis

Subdomain: Probability and conditional probability

Difficulty: Medium

Question 14

The exponential function f is defined by $f(x) = ab^x$, where a and b are positive constants. If $f(2) = k$, which of the following equivalent expressions explicitly displays the value of the parameter k as a constant base or coefficient?

- A. $f(x) = ab^2 \cdot b^{x-2}$
- B. $f(x) = k \cdot b^{x-2}$
- C. $f(x) = k \cdot b^x$
- D. $f(x) = (k / b^2) \cdot b^x$

Answer: B

Domain: Advanced Math

Subdomain: Nonlinear functions

Difficulty: Medium

Question 15

Triangle PQR is inscribed inside a circle. The minor arc PQ is congruent to the minor arc QR. If the interior degree measure of inscribed angle Q is equal to 44° , what is the interior degree measure of angle R?

- A. 44°
- B. 68°
- C. 88°
- D. 136°

Answer: B

Domain: Geometry and Trigonometry

Subdomain: Circles

Difficulty: Difficult

Question 16

The cumulative frequency table below tracks the daily commute times of 50 urban transit workers.

Commute Time	Cumulative Number of Workers
10 to 20 minutes	8
10 to 30 minutes	17
10 to 40 minutes	35
10 to 50 minutes	46
10 to 60 minutes	50

If one of these transit workers is selected completely at random, what is the probability that the chosen worker has a daily commute time between 31 and 40 minutes?

- A. 9/50
- B. 17/50
- C. 18/50
- D. 35/50

Answer: C

Domain: Problem-Solving and Data Analysis

Subdomain: One-variable data: distributions and measures of center and spread

Difficulty: Medium

Question 17

The system of equations below has exactly one unique real solution (x, y) in the standard xy -plane:

$$y = cx^2 - 6x + 5$$

$$y = 2x - 3$$

where c is a constant. What is the value of c ?

- A. 1
- B. 2
- C. 4
- D. 8

Answer: B

Domain: Advanced Math

Subdomain: Systems of equations in 2 variables

Difficulty: Medium

Question 18

The polynomial equation $y = x^3 + bx^2 + cx - 72$ has exactly two distinct positive integer roots, and one of these roots is located at $x = 6$. What is the value of the linear coefficient constant c ?

Answer: 60

Domain: Advanced Math

Subdomain: Nonlinear functions

Difficulty: Difficult

Question 19

In the standard xy -plane, the vertex of the parabola $y = -3(x - 5)^2 + k$ is located at the point $(5, 22)$. What is the y -intercept of this parabola?

Answer: -53

Domain: Advanced Math

Subdomain: Nonlinear functions

Difficulty: Easy

Question 20

The data sets X and Y each consist of 30 distinct whole numbers. The data values in Set X have a range of 60 and a standard deviation of 11.4. Set Y is created by multiplying every individual data value in Set X by exactly 4 and then adding 15 to each resulting value. What is the standard deviation of data Set Y?

Answer: 45.6

Domain: Problem-Solving and Data Analysis

Subdomain: One-variable data: distributions and measures of center and spread

Difficulty: Medium

Question 21

The expression below can be simplified to the form $ax^m y^n$, where a, m, and n are integers.

$$(5x^2 \cdot y^{-4} \cdot x^3)^2 / (x^{-2} \cdot y^3)$$

What is the value of $m + n$?

Answer: 1

Domain: Advanced Math

Subdomain: Equivalent expressions

Difficulty: Easy

Question 22

An isosceles triangle is inscribed inside a circle. The triangle has side lengths 13 inches, 13 inches, and 12 inches. What is the radius length of the circumscribed circle, expressed as a decimal?

Answer: 7.5

Domain: Geometry and Trigonometry

Subdomain: Lines, angles, and triangles

Difficulty: Difficult

Module 3:

Question 1

A boutique owner increases the retail price of a luxury handbag by $p\%$. After a month without a sale, the owner applies a $q\%$ discount to the new price, which brings the final price of the handbag back to exactly its original starting price. Which of the following correctly expresses q in terms of p ?

- A. $q = p$
- B. $q = 100p / (100 - p)$
- C. $q = 100p / (100 + p)$
- D. $q = p / (100 + p)$

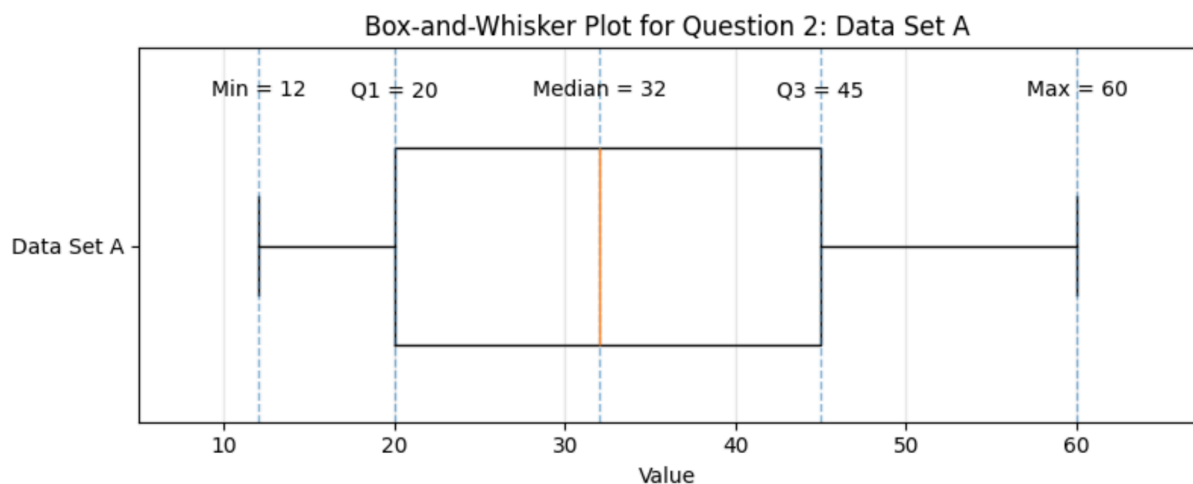
Answer: C

Domain: Problem-Solving and Data Analysis

Subdomain: Percentages

Difficulty: Medium

Question 2



The box-and-whisker plot for Data Set A is shown above. A computer algorithm generates Data Set B by applying the transformation $y = -3x + 15$ to every single value in Data Set A. What is the interquartile range (IQR) of Data Set B?

- A. -75
- B. 25
- C. 75
- D. 120

Answer: C

Domain: Problem-Solving and Data Analysis

Subdomain: One-variable data: distributions and measures of center and spread

Difficulty: Medium

Question 3

Set K consists of n consecutive integers. If the median of the numbers in Set K is NOT an integer, which of the following must be true?

- I. n is an even number.
- II. The mean of the numbers in Set K is not an integer.
- III. The sum of the numbers in Set K is an integer.

- (A) I only
- (B) I and II only
- (C) I and III only
- (D) I, II, and III

Answer: D

Domain: Problem-Solving and Data Analysis

Subdomain: One-variable data: distributions and measures of center and spread

Difficulty: Difficult

Question 4

A chemical distributor blends two types of cleaning solvent: Standard and Industrial. Standard solvent costs c dollars per liter, and Industrial solvent costs d dollars per liter. The distributor creates a 500-liter mixture by blending x liters of Standard and y liters of Industrial solvent. If the total cost of the mixture is T dollars, which of the following expressions correctly isolates x , the volume of Standard solvent used?

- A. $(T - 500d) / (c - d)$
- B. $(T - 500c) / (c - d)$
- C. $(500d - T) / (c + d)$
- D. $T / (c + d) - 500$

Answer: A

Domain: Algebra

Subdomain: Systems of 2 linear equations in 2 variables

Difficulty: Difficult

Question 5

A radioactive isotope decays according to the model $M(t) = M_0(0.5)^{t/h}$, where M_0 is the initial mass, t is time in years, and h is the half-life in years. If the mass of the isotope decreases by exactly 36% every 10 years, which of the following equations can be used to solve for the half-life h ?

- A. $0.36 = (0.5)^{10/h}$
- B. $0.64 = (0.5)^{10/h}$
- C. $0.64 = (0.5)^{h/10}$
- D. $1.36 = (0.5)^{10/h}$

Answer: B

Domain: Advanced Math

Subdomain: Nonlinear functions

Difficulty: Medium

Question 6

The quadratic function $f(x) = ax^2 + bx + c$ has a vertex located on the line $y = x$. If the

x-coordinate of the vertex is $6/a$, which of the following correctly expresses the constant c in terms of a and b ?

- A. $c = 6/a - 36/a$
- B. $c = (6 + b^2) / 4a$
- C. $c = (b^2 + 24) / 4a$
- D. $c = (6a + 36) / a$

Answer: C

Domain: Advanced Math

Subdomain: Nonlinear functions

Difficulty: Difficult

Question 7

The continuous polynomial function $y = p(x)$ has exactly three x-intercepts located at $x = -4$, $x = 2$, and $x = 9$. A new function is defined as $g(x) = p(-2x + 5)$. What is the sum of the x-intercepts of the graph of $y = g(x)$?

- A. -3.5
- B. 3.5
- C. 4.0
- D. 7.0

Answer: C

Domain: Advanced Math

Subdomain: Nonlinear functions

Difficulty: Difficult

Question 8

In the xy -plane, the equation $x^2 + y^2 - 6kx + 10ky + 9k^2 = 0$ represents a circle, where k is a positive constant. What is the area of this circle in terms of k ?

- A. $9\pi k^2$
- B. $16\pi k^2$
- C. $25\pi k^2$
- D. $34\pi k^2$

Answer: C

Domain: Geometry and Trigonometry

Subdomain: Circles

Difficulty: Difficult

Question 9

A solid glass right square pyramid is sliced by a plane parallel to its base. This slice divides the original pyramid into a smaller pyramid at the top and a frustum at the bottom. If the volume of the smaller top pyramid is exactly $1/64$ the volume of the original full pyramid, what is the ratio of the height of the top pyramid to the height of the bottom frustum?

- A. 1 : 3
- B. 1 : 4
- C. 1 : 8
- D. 1 : 63

Answer: A

Domain: Geometry and Trigonometry

Subdomain: Area and volume formulas

Difficulty: Difficult

Question 10

For all $x \neq 4$, the expression below is equivalent to $3x - p$, where c and p are constants.

$$(3x^2 - 14x + c) / (x - 4)$$

What is the value of $c + p$?

- A. -6
- B. -2
- C. 6
- D. 10

Answer: D

Domain: Advanced Math

Subdomain: Equivalent expressions

Difficulty: Medium

Question 11

A rectangular holding tank has an internal length of $3x$ meters, a width of $2x$ meters, and a height of x meters. The fluid inside this tank is completely drained and pumped into a new empty tank shaped like a regular square pyramid. If the pyramid has a base edge length of $2x$ meters, what must be the height of the pyramid in terms of x so that it is filled to its exact maximum capacity?

- A. $4.5x$
- B. $9x$
- C. $13.5x$
- D. $18x$

Answer: A

Domain: Geometry and Trigonometry

Subdomain: Area and volume formulas

Difficulty: Difficult

Question 12

The system of equations below has infinitely many solutions:

$$4x - cy = 18$$

$$dx + 7y = -27$$

What is the value of the product cd ?

- A. -42
- B. -28
- C. 28
- D. 42

Answer: B

Domain: Algebra

Subdomain: Systems of 2 linear equations in 2 variables

Difficulty: Difficult

Question 13

A circle in the xy -plane is given by the equation $(x - 5)^2 + (y + 1)^2 = 34$. A line is drawn tangent to this circle at the coordinate point $(8, 4)$. If the equation of this tangent line is written in the form $y = mx + b$, what is the value of the y -intercept, b ?

- A. -4.5
- B. 8.8
- C. 12.5
- D. 16.0

Answer: B

Domain: Geometry and Trigonometry

Subdomain: Circles

Difficulty: Difficult

Question 14

In a medical trial, Solution X contains 30% active compound and Solution Y contains 80% active compound. A technician mixes an unknown volume of Solution X with exactly 15 liters of Solution Y to yield a final mixture that contains exactly 60% active compound. What fraction of the total volume of the final mixture is made up of Solution X?

- A. $\frac{2}{5}$
- B. $\frac{3}{5}$
- C. $\frac{2}{3}$
- D. $\frac{3}{4}$

Answer: A

Domain: Problem-Solving and Data Analysis

Subdomain: Percentages

Difficulty: Difficult

Question 15

The system of equations $y = \frac{1}{2}x + 4$ and $y = bx - 3$ intersects in the first quadrant of the xy -plane. Which of the following is a possible value for b ?

- (A) -0.5
- (B) 0.2

- (C) 0.5
(D) 1.2

Answer: D

Domain: Advanced Math

Subdomain: Nonlinear functions

Difficulty: Medium

Question 16

The quadratic function $h(x)$ is defined as $h(x) = -2(x - 5)^2 + 20$. For how many integer values of k does the equation $|h(x)| = k$ have exactly 4 distinct real solutions?

- A. 0
B. 19
C. 20
D. Infinite

Answer: B

Domain: Advanced Math

Subdomain: Nonlinear equations in 1 variable

Difficulty: Difficult

Question 17

A data set consists of 7 unique positive integers. The median of the set is 14, and the maximum value in the set is 30. What is the absolute minimum possible value for the mean of this data set?

- A. 9.00
B. 10.28
C. 11.57
D. 14.00

Answer: C

Domain: Problem-Solving and Data Analysis

Subdomain: One-variable data: distributions and measures of center and spread

Difficulty: Difficult

Question 18

The roots of the equation $2x^2 - 16x + c = 0$ are r_1 and r_2 . If $(r_1)^2 + (r_2)^2 = 34$, what is the value of the constant c ?

Answer: 30

Domain: Advanced Math

Subdomain: Nonlinear equations in 1 variable

Difficulty: Medium

Question 19

A student takes a standardized test with 60 questions. The scoring system awards 5 points for a correct answer, deducts 2 points for an incorrect answer, and awards 0 points for

questions left blank. The student scored a total of 131 points and left exactly 3 questions blank. How many questions did the student answer correctly?

Answer: 35

Domain: Algebra

Subdomain: Systems of 2 linear equations in 2 variables

Difficulty: Medium

Question 20

In the standard xy -plane, the circle defined by the equation $x^2 + y^2 - 12x + 14y + c = 0$ has a total circumference of 16π . What is the value of the constant c ?

Answer: 21

Domain: Geometry and Trigonometry

Subdomain: Circles

Difficulty: Medium

Question 21

A solid metal cylinder has a base radius of r and a height of h . The cylinder is melted down and recast. In the new cylinder, the radius is increased by 20%, and the height is decreased by $p\%$. If the total volume of the cylinder remains exactly unchanged, what is the value of p ? Ignore the percent sign when gridding your answer.

Answer: 275/9

Domain: Geometry and Trigonometry

Subdomain: Area and volume formulas

Difficulty: Difficult

Question 22

The function $f(x)$ is defined as $f(x) = x^2 - 8x + 19$. A new function $g(x)$ is formed by transforming $f(x)$ such that $g(x) = f(x - 4) + 5$. If the vertex of $g(x)$ is located at the point (h, k) , what is the value of $h + k$?

Answer: 16

Domain: Advanced Math

Subdomain: Nonlinear functions

Difficulty: Medium

